Positive-Data Learning of Multiple Context-Free Languages under Fixed Finite-Semigroup Typing

Takayuki Kuriyama
Independent Researcher, Tokyo, Japan
growup.kuriyama@gmail.com

Abstract

Fix a surjective finite-semigroup morphism $s : \Sigma^+ \to \mathsf{S}$ and a fan-out bound f. We study positive-data learning of multiple context-free languages under componentwise s-typed substitutability. A canonical fresh-identity completion $\widehat{s} : \Sigma^* \to \mathsf{S}^\sharp$ evaluates grammar yields, with empty words treated separately. We specialize Yoshinaka's signature-based positive-data learner to fixed finite-semigroup typing, make the set-driven reconstruction explicit, and analyze characteristic samples beyond their cardinality. At bounded rule rank the hypothesis updates are polynomial-time.

Our main quantitative measure is the total length of characteristic positive data relative to the size $\|G\|$ and original rule thickness θ_G of a reduced good MCFG presentation. At fixed fan-out and rank, boundary-window typings admit $O(\|G\|^2\bar{\theta}_G)$ characteristic data; combining a window with any fixed finite group yields $O(\|G\|^3\bar{\theta}_G)$, including noncommutative groups. We refine the abelian constants via zero-sum theory and extend the cubic bound to typings with positive-image semigroup in the pseudovariety **LG** of local groups. For rank-one rules, every fixed finite typing has a grammar-size polynomial bound. A distinct language-based condition gives polynomial data for abelian-filtered targets with bounded-rank good presentations, even when the positive-image semigroup lies outside **LG**.

We provide a computable rule-only witness budget and a separate distributional-representative upper bound. At the pseudovariety level, the local-group upper boundary has lower-bound obstructions at fan-out one. We also give a boundary-typed non-context-free example with intrinsic rule rank two. The general higher-fan-out superpolynomial data lower-bound problem remains open.

Keywords: grammatical inference; multiple context-free grammars; positive data; finite-semigroup typing; multidimensional substitutability; Gold identification; rule rank.

1 Introduction

Positive-data grammatical inference asks when a language can be recovered from an enumeration of its positive examples alone [1, 2]. Distributional approaches make this possible for restricted grammar classes by turning observed interchangeability of substrings or tuples into grammar nonterminals. For context-free grammars (CFGs) this includes substitutable languages [3] and Yoshinaka's (k, ℓ) hierarchy [12]. Yoshinaka extended the idea to multiple context-free grammars (MCFGs), whose nonterminals generate tuples and whose substitutions are tested by multi-hole sentence contexts [13,14]. MCFGs are a standard mildly context-sensitive formalism for discontinuous and cross-serial dependencies [30, 31].

This paper fixes, as part of the learning bias, a surjective finite-semigroup homomorphism $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ and compares observed *nonempty* tuple components only when their s-values agree. The canonical fresh-identity completion $\widehat{s} : \Sigma^* \to \mathsf{S}^\sharp$ is used only to evaluate grammar templates, including empty terminal gaps. The typing and the fan-out bound are fixed before any data are seen, separating the choice of a finite bias from learning under that bias [5]; prefix–suffix windows and DFA transition morphisms are typical instances. Other biases restrict derivations by regular control sets [6, 7] or restrict the evidence used to compare factors [9, 11]; positive-data learning of other mildly context-sensitive formalisms is studied in [19], and MCFG learning with membership queries or dual context sets in [16, 18, 20]. A recent substitution lemma for MCFLs [23] is a necessary condition for membership, not a learning criterion.

Prior work and what is not claimed. The qualitative learning mechanism is not new. The reconstruction operator of Section 3 is Yoshinaka's observed-tuple grammar [14, Section 4.1] with two changes: substitution links are guarded by equality of componentwise types, and ε is admitted by a separate start rule. Its soundness and completeness proofs are typed transcriptions of [14, Lemmas 3–5], and the unbounded-rank variant is the typed form of [14, Corollary 2]. At bounded rule rank the typed class is, moreover, the substitutable class of a finite sort refinement in Yoshinaka's signature framework (Proposition 3.17), so identifiability, polynomial updates and characteristic sets of bounded *cardinality* follow from [15, Theorem 1 and Lemma 3]; that theorem is proved there for the context-indexed construction $\mathrm{SUBST}_{\mathrm{D}}$, and Lemma 3.18 relates our operator to the substructure-indexed $\mathrm{SUBST}_{\mathrm{P}}$. Polynomial characteristic data for fixed prefix–suffix typing of CFGs is due to Yoshinaka [12]; the companion CFG paper [25] (arXiv v5, with a frozen TCS R1 source) treats fan-out one, including a window–group bound and an operator-independent semilattice lower bound. The non-branching lower bound used in Section 5 is due to Gallot [21].

Contributions. Our object is the *total encoded length* of characteristic positive data relative to an MCFG presentation, for typed *tuples* at higher rule rank, where several component boundaries must be preserved simultaneously. For a reduced good MCFG G put $\bar{\theta}_G = \max\{1, \theta_G\}$, where θ_G is rule thickness (Section 4). With alphabet, fan-out, rule-rank bound and typing fixed:

Fixed typing regime	Positive-data length	Source
Rule rank one; arbitrary typing	$\mathrm{poly}(\|G_{\mathrm{in}}\|)$	Thm. 4.5
Prefix–suffix window	$O(\|G\|^2 \bar{\theta}_G)$	Thm. 4.9
Window and any finite group	$O(\|G\|^3 \bar{\theta}_G)$	Thm. C.7
Positive image in **LG**	$O(\|G\|^3 \bar{\theta}_G)$	Thm. D.7

The group result includes noncommutative groups; the abelian case has sharper constants (Theorem B.4). The **LG** hypothesis concerns the positive-image semigroup; its known algebraic characterization [37] is distinct from the learning bound, and the matching pseudovariety-level obstruction is established only at fan-out one (Corollary D.9). We also give a computable rule-only data budget (Proposition 4.15), a language-based bound for abelian-filtered targets (Theorem 4.19), and a boundary-typed non-context-free target of intrinsic fan-out and rule rank two (Theorem 5.3). Characteristic data concern language-level exactness of a set-driven operator; syntactic stabilization is supplied by a separate conservative learner.

Relationship to the public preprint. An earlier, differently scoped draft was circulated as arXiv:2605.11644. The present manuscript substantially revises its formulation and states only the results proved here; the two should not be treated as textually identical versions.

Section 2 fixes the model, Section 3 the reconstruction and its exactness, Section 4 the quantitative bounds, and Section 5 the separations.

2 Preliminaries

Throughout, Σ is a finite alphabet, ε the empty word, and $\Sigma^+ := \Sigma^* \setminus \{\varepsilon\}$. If $\Sigma = \emptyset$, the only languages are $\emptyset$ and $\{\varepsilon\}$ and are handled directly.

2.1 Positive-Data Learning

Definition 2.1 (Text, TxtEx identification, conservativeness)**.** Fix a pause symbol $\perp \notin \Sigma$. A *text* for $L \subseteq \Sigma^*$ is an infinite sequence over $L \cup \{\perp\}$ in which every element of L occurs; thus $\perp, \perp, \ldots$ is the text for $\emptyset$. The *content* of a finite prefix is the set of elements of Σ^* occurring in it. A *learner* is a computable map from finite prefixes to grammars. It *TxtEx-identifies* L [1] if on every text for L its outputs are, from some stage on, one fixed grammar H with $L(H) = L$, and it TxtEx-identifies a class if it does so for each member. A learner is *conservative* if it changes its hypothesis only on a datum $w \in \Sigma^*$ not generated by the current hypothesis. We fix a hypothesis H_0 with $L(H_0) = \emptyset$.

Definition 2.2 (Set-driven operator and characteristic sample)**.** A *set-driven reconstruction operator* $\mathcal{B}$ maps each finite $K \subseteq \Sigma^*$ to a grammar $\mathcal{B}(K)$ depending only on the set K; it is *sample-consistent* if $K \subseteq L(\mathcal{B}(K))$. For a language L, a finite $K^\star \subseteq L$ is a *characteristic sample for* $\mathcal{B}$ if $K^\star \subseteq K \subseteq L$ implies $L(\mathcal{B}(K)) = L$ for every finite K (cf. [4]). Unlike TxtEx identification this is a statement about generated languages: $\mathcal{B}(K)$ need not stabilize syntactically as K grows.

For finite $K \subseteq \Sigma^*$ put $\|K\|_+ := \sum_{w \in K}(|w| + 1)$, and for a tuple $\vec{u} = (u_1, \ldots, u_d)$ put $\|\vec{u}\|_1 := \sum_i |u_i|$. Every polynomial bound below states whether it concerns cardinality or $\|\cdot\|_+$.

2.2 Fixed Finite-Semigroup Typing

Following the companion CFG paper [25], the learning bias is a surjective semigroup morphism $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ onto a finite semigroup, given by its multiplication table and the letter images (so $s(w)$ is computable in linear time). A finite codomain can always be replaced by the positive image, so surjectivity loses no generality.

Definition 2.3 (Fresh completion, tuple types, refinement)**.** Let $\mathsf{S}^\sharp = \mathsf{S} \uplus \{\mathbf{1}_\sharp\}$ be obtained by adjoining a *new* identity (even if S has one), and let $h = \widehat{s} : \Sigma^* \to M := \mathsf{S}^\sharp$ be the monoid morphism with $\widehat{s}(\varepsilon) = \mathbf{1}_\sharp$ and $\widehat{s}|_{\Sigma^+} = s$. For $\vec{w} = (w_1, \ldots, w_d)$ put $h^{(d)}(\vec{w}) = (h(w_1), \ldots, h(w_d))$; on $(\Sigma^+)^d$ this is the semigroup tuple type $s^{(d)}$. The value $\mathbf{1}_\sharp$ only evaluates empty template factors; it is not a type of the bias. For typings s, t on Σ^+ write $s \preceq t$ if $\ker t \subseteq \ker s$ on Σ^+ (equivalently, s factors through the positive image of t), and let (s, t) denote $w \mapsto (s(w), t(w))$ restricted to its positive image.

Monoid implementations. A finite-monoid morphism $h : \Sigma^* \to M$ is used as a presentation of the typing $h|_{\Sigma^+}$ onto its positive image; accordingly $h_0 \preceq h_1$ means $h_0|_{\Sigma^+} \preceq h_1|_{\Sigma^+}$, products of implementations are formed componentwise, and $\mathcal{C}^{\mathrm{mcf}}_{f,h}$, $\mathcal{B}_{f,h}$, $\mathcal{A}_{f,h}$ mean $\mathcal{C}^{\mathrm{mcf}}_{f,h|_{\Sigma^+}}$, $\mathcal{B}_{f,\widehat{s}}$, $\mathcal{A}_{f,\widehat{s}}$ with $s = h|_{\Sigma^+}$. Lemma 3.7 shows that nothing below depends on the choice of implementation. Bounds in terms of $|M|$ refer to $M = \mathsf{S}^\sharp$, so $|M| = |\mathsf{S}| + 1$; algebraic conditions (such as **LG**) always refer to the positive image S. The trivial typing s_{triv} has one positive type and is implemented by $h_{\mathrm{triv}} : \Sigma^* \to \{1\}$.

Definition 2.4 (Prefix–suffix typing)**.** For $k, \ell \geq 0$ put $m_0 := \max\{1, k+\ell\}$, and for $|w| \geq m_0$ let $\mathrm{pref}_k(w)$, $\mathrm{suf}_\ell(w)$ be the prefix and suffix of the indicated lengths (length-zero windows are ε). Define

$$h_\Sigma^{k,\ell}(w) = \begin{cases} w, & |w| < m_0, \\ \langle \mathrm{pref}_k(w), \mathrm{suf}_\ell(w) \rangle, & |w| \geq m_0, \end{cases}$$

with disjoint tags for short words and pairs, and put $M_{k,\ell} := h_\Sigma^{k,\ell}(\Sigma^*)$. For short x, y and pairs $\langle u, v\rangle, \langle u', v'\rangle$ define

$$x \cdot y := h_\Sigma^{k,\ell}(xy), \qquad x \cdot \langle u, v\rangle := \langle \mathrm{pref}_k(xu), v\rangle,$$
$$\langle u, v\rangle \cdot y := \langle u, \mathrm{suf}_\ell(vy)\rangle, \quad \langle u, v\rangle \cdot \langle u', v'\rangle := \langle u, v'\rangle.$$

A case check gives $h_\Sigma^{k,\ell}(xy) = h_\Sigma^{k,\ell}(x) \cdot h_\Sigma^{k,\ell}(y)$; since $h_\Sigma^{k,\ell}$ is onto $M_{k,\ell}$, associativity is inherited from concatenation and $h_\Sigma^{k,\ell}(\varepsilon)$ is the identity. Thus $h_\Sigma^{k,\ell}$ is a finite-monoid morphism; its positive restriction $s_\Sigma^{k,\ell}$ is the boundary typing, and since the short-word tag separates ε from Σ^+, $h_\Sigma^{k,\ell}$ is isomorphic to $\widehat{s_\Sigma^{k,\ell}}$. A typing h with $h \preceq h_\Sigma^{k,\ell}$ is called (k,ℓ)-*boundary-determined*, or briefly a *boundary typing*.

2.3 Multiple Context-Free Grammars

Definition 2.5 (Standard MCFG presentation)**.** An MCFG [30, 31] is $G = (V, \Sigma, P, S, \mathrm{fan})$ with finite V, $S \in V$, $\mathrm{fan}(S) = 1$ and $\mathrm{fan}(A) \geq 1$. A rule of rank $r \geq 0$ is $A \to \rho(B_1, \ldots, B_r)$ with template $\rho = (\alpha_1, \ldots, \alpha_{d_0})$, $d_0 = \mathrm{fan}(A)$, $d_j = \mathrm{fan}(B_j)$, where each α_i is a word over $\Gamma_\rho := \Sigma \uplus \Xi_1 \uplus \cdots \uplus \Xi_r$, $\Xi_j := \{\xi_j^1, \ldots, \xi_j^{d_j}\}$, and every variable occurs at most once in $\alpha_1 \cdots \alpha_{d_0}$ (*linearity*); the rule is *nondeleting* if every variable occurs exactly once (derived components may still be empty). For $\vec{u}_j \in (\Sigma^*)^{d_j}$, $\rho(\vec{u}_1, \ldots, \vec{u}_r)$ substitutes $u_{j,\kappa}$ for ξ_j^κ; for $r = 0$ the rule is a constant rule. The tuple languages $L_A(G)$ form the least family closed under the rules, and $L(G) := \{w : (w) \in L_S(G)\}$. We write $A \Rightarrow_G^* \vec{u}$ for a *terminal derivation* (a finite derivation tree rooted at A evaluated bottom-up); one from S is *successful*. G is an f-MCFG if $\mathrm{fan}(A) \leq f$ for all A, and has rule-rank cap $r_{\max}$ if all ranks are at most $r_{\max}$.

We write $\mathsf{MCFL}(f, r)$ for the languages of MCFGs of fan-out $\leq f$ and rule rank $\leq r$, so that f-MCFL $= \bigcup_{r<\infty} \mathsf{MCFL}(f, r)$. Minimum rule rank of a language refers to all presentations, not to a given grammar. $\|G\|$ is the length of a fixed symbol-level encoding counting nonterminal and rule records and every terminal or variable occurrence in templates. For nonterminals of equal fan-out d, $A \to B$ abbreviates the identity-template rule $A \to (\xi_1^1, \ldots, \xi_1^d)(B)$, and $A \to \vec{c}$ a constant rule. Rule variables are always ξ_j^κ; terminals in examples are in `typewriter` font.

A nondeleting rule is *nonpermuting* if, for each j, $\xi_j^1, \ldots, \xi_j^{d_j}$ occur in this order in $\alpha_1 \cdots \alpha_{d_0}$; following [14, Section 2.3] it is λ-*free* if no α_i is empty, and a nonpermuting rule is *nonmerging* if no α_i contains a factor $\xi_j^\kappa \xi_j^{\kappa+1}$. A rule, template or presentation is *good* if it is λ-free, nondeleting, nonpermuting and nonmerging. By induction on derivations, every component of every tuple derived from a nonterminal of a λ-free grammar is nonempty. A nonterminal is *reachable* if it is reachable from S in the graph with an edge $A \to B$ whenever B is a child in a rule for A, and *productive* if $L_A(G) \neq \emptyset$; G is *reduced* if all nonterminals are both.

Proposition 2.6 (Good normalization with rank preservation)**.** *Let G_{in} be a finite f-MCFG of rule rank at most $r_{\max}$. There is a finite good f-MCFG G_{good} of rule rank at most $r_{\max}$ with $L(G_{\mathrm{good}}) = L(G_{\mathrm{in}}) \setminus \{\varepsilon\}$ and, for fixed $f, r_{\max}$, $\|G_{\mathrm{good}}\| = O_{f,r_{\max}}(\|G_{\mathrm{in}}\|)$.*

Proof. Trim; apply the nondeleting construction (f3) of [30, proof of Lemma 2.2] by retaining subsets of output components; decorate nonterminals by component permutations; decorate them by realizable nonempty-component patterns and project away empty components (this removes ε and makes all templates λ-free); finally apply Yoshinaka's nonmerging conversion [14, Section 2.3, Lemma 1]. No step increases fan-out or rule rank, and with $f, r_{\max}$ fixed each step creates $O_{f,r_{\max}}(1)$ copies of each nonterminal and rule (at most 2^f, $f!$ and $2^{fr_{\max}}$ decorations). The derivation correspondences and the size bookkeeping are given in Appendix A; cf. also [32, Section 2.1] and [14, Section 2.3, after Example 2]. □

The proposition controls size, fan-out and rank, not thickness; thickness transfer for reduced λ-free inputs is Proposition A.1. Fan-out and rule rank are distinct resources: there is no rank reduction preserving an arbitrary fixed fan-out [24, 33]. The reconstruction below therefore imposes neither a binary normal form nor a known rank cap.

2.4 Tuple Contexts and Substitutability

A *sentence context of arity* d is a word $E = v_0\Box_1 v_1 \cdots v_{d-1}\Box_d v_d$ with each hole occurring once, $v_0, v_d \in \Sigma^*$ and $v_1, \ldots, v_{d-1} \in \Sigma^+$; such contexts are *admissible*, and $\mathcal{E}_d$ denotes their set ([14, Section 3.1]). $E[\vec{w}]$ fills $\Box_i$ by w_i, $|E|_{\text{fix}} := \sum_{i=0}^{d} |v_i|$, and E *accepts* $\vec{w}$ in L if $E[\vec{w}] \in L$.

Definition 2.7 (Tuple occurrence)**.** For $w = a_1 \cdots a_n$ number the cut positions $⓪a_1① \cdots a_n⓷$ and write $w[i : i'] := a_{i+1} \cdots a_{i'}$. An *arity-$d$ tuple occurrence* in w is a sequence $0 \le i_1 < i'_1 < i_2 < i'_2 < \cdots < i_d < i'_d \le n$; it determines $\vec{x} = (w[i_1 : i'_1], \ldots, w[i_d : i'_d]) \in (\Sigma^+)^d$ and the context $E \in \mathcal{E}_d$ obtained by replacing these intervals by $\Box_1, \ldots, \Box_d$, so that $E[\vec{x}] = w$. For a finite K, $\mathrm{Obs}_d(K)$ is the set of tuples $\vec{x}$ arising from occurrences in words of K, i.e. $\vec{x} \in (\Sigma^+)^d$ with $E[\vec{x}] \in K$ for some $E \in \mathcal{E}_d$, and $\mathrm{Obs}(K) := \bigcup_{d \le f} \mathrm{Obs}_d(K)$.

For $L \subseteq \Sigma^*$ and $\vec{x} \in (\Sigma^*)^d$ the *d-tuple distribution* is $\mathcal{D}_L^{(d)}(\vec{x}) := \{E \in \mathcal{E}_d : E[\vec{x}] \in L\}$.

Definition 2.8 ((f,s)-tuple substitutability)**.** Let $f \ge 1$ and $s : \Sigma^+ \to \mathsf{S}$ a finite-semigroup morphism. A language $L \subseteq \Sigma^*$ is *(f,s)-tuple-substitutable* if for every $1 \le d \le f$ and all $\vec{x}, \vec{y} \in (\Sigma^+)^d$,

$$s^{(d)}(\vec{x}) = s^{(d)}(\vec{y}) \ \text{ and } \ \mathcal{D}_L^{(d)}(\vec{x}) \cap \mathcal{D}_L^{(d)}(\vec{y}) \neq \emptyset \implies \mathcal{D}_L^{(d)}(\vec{x}) = \mathcal{D}_L^{(d)}(\vec{y}).$$

Empty components are never compared (Yoshinaka's convention [14, Section 3.1 and footnote 3]; see [8] for CFGs); $\varepsilon \in L$ is handled at the start symbol. The restriction matters already at fan-out one: $\mathtt{a}$ and ε share a context in $\mathtt{a}^+$ but have different distributions. At fan-out one, $\mathcal{D}_L^{(1)}(x)$ is the two-sided distribution $\{(u,v) : uxv \in L\}$. For $\vec{x}, \vec{y} \in (\Sigma^+)^d$ write $\vec{x} \equiv_{L,h}^{d} \vec{y}$ (or $\vec{x} \equiv_{L,h} \vec{y}$) if $h^{(d)}(\vec{x}) = h^{(d)}(\vec{y})$ and $\mathcal{D}_L^{(d)}(\vec{x}) = \mathcal{D}_L^{(d)}(\vec{y})$; this is an equivalence relation.

Definition 2.9 (The target class)**.** $\mathcal{C}^{\mathrm{mcf}}_{f,s}$ is the class of (f,s)-tuple-substitutable f-MCFLs. A target in $\mathcal{C}^{\mathrm{mcf}}_{f,h}$ for a boundary typing h is called *boundary-typed*.

If $h \preceq h'$ then $\mathcal{C}^{\mathrm{mcf}}_{f,h} \subseteq \mathcal{C}^{\mathrm{mcf}}_{f,h'}$, since equal finer types are equal coarser types; in particular $\mathcal{C}^{\mathrm{mcf}}_{f,h_{\mathrm{triv}}} \subseteq \mathcal{C}^{\mathrm{mcf}}_{f,h}$. The class is not monotone in f: only $\mathcal{C}^{\mathrm{mcf}}_{f+1,h} \cap f\text{-MCFL} \subseteq \mathcal{C}^{\mathrm{mcf}}_{f,h}$ holds, and already for trivial typing $\{w\#w^R\}$ is 1- but not 2-dimensionally substitutable [14, Section 3.1, Example 5]; Yoshinaka notes that $\mathsf{SL}(p,r)$ and $\mathsf{SL}(p+1,r)$ are incomparable for $r \ge 1$ [14, Section 4.1]. Thus f is part of the bias. For every $\mathsf{Acc} \subseteq M$, $h^{-1}(\mathsf{Acc}) \in \mathcal{C}^{\mathrm{mcf}}_{f,h}$: it is regular, and equal componentwise types give equal total types in every context.

Proposition 2.10 (Intersection and abelian filtering)**.** *Fix f and h.*

(i) If L_1, L_2 are (f,h)-tuple-substitutable, so is $L_1 \cap L_2$.

(ii) If L is (f,h)-tuple-substitutable, $\mathbb{A}$ is an abelian group, $\varphi : \Sigma^* \to \mathbb{A}$ a monoid morphism, $\mathcal{H} \leq \mathbb{A}$ and $\gamma \in \mathbb{A}$, then $L \cap \varphi^{-1}(\gamma + \mathcal{H})$ is (f,h)-tuple-substitutable.

Whenever the intersection is an f-MCFL, it belongs to $\mathcal{C}^{\mathrm{mcf}}_{f,h}$.

Proof. (i) A context accepting equal-type $\vec{x}, \vec{y}$ in $L_1 \cap L_2$ accepts both in each L_i, so their distributions agree in each L_i, hence in the intersection. (ii) If a context accepts $\vec{x}, \vec{y}$ in $\varphi^{-1}(\gamma + \mathcal{H})$, then $\delta := \sum_i (\varphi(y_i) - \varphi(x_i)) \in \mathcal{H}$; by commutativity $\varphi(E'[\vec{y}]) - \varphi(E'[\vec{x}]) = \delta$ for every E', so coset membership transfers in both directions. Thus $\varphi^{-1}(\gamma + \mathcal{H})$ is (f, h_{triv})-, hence (f,h)-tuple-substitutable; apply (i). □

Consequently every f-MCFL of the form $h^{-1}(\mathsf{Acc}) \cap \varphi^{-1}(\gamma + \mathcal{H})$ lies in $\mathcal{C}^{\mathrm{mcf}}_{f,h}$; such factorizations are not supplied to the learner.

All interaction between templates and contexts below rests on one observation.

Lemma 2.11 (Induced child contexts)**.** *Let $\rho = (\alpha_1, \ldots, \alpha_e)$ be a good template of rank r with child arities $d_1, \ldots, d_r$, let $E \in \mathcal{E}_e$, let $1 \leq j \leq r$, and fix $\vec{u}_i \in (\Sigma^+)^{d_i}$ for $i \neq j$. Let E_j be the word obtained from $E[(\alpha_1, \ldots, \alpha_e)]$ by replacing ξ^κ_j by $\square_\kappa$ and every other variable ξ^ι_i by $u_{i,\iota}$. Then $E_j \in \mathcal{E}_{d_j}$ and $E_j[\vec{x}] = E[\rho(\vec{u}_1, \ldots, \vec{u}_{j-1}, \vec{x}, \vec{u}_{j+1}, \ldots, \vec{u}_r)]$ for every $\vec{x} \in (\Sigma^*)^{d_j}$.*

Proof. The identity holds because substitutions commute. By linearity and nondeletion each $\square_\kappa$ occurs exactly once in E_j, and by nonpermutation in the order $\kappa = 1, \ldots, d_j$, since E places $\alpha_1, \ldots, \alpha_e$ from left to right. Consider ξ^κ_j and $\xi^{\kappa+1}_j$. If both lie in one α_i, nonmerging puts a terminal or a variable ξ^ι_i with $i \neq j$ between them, which becomes a nonempty word. Otherwise they lie in $\alpha_i, \alpha_{i'}$ with $i < i'$, and the material between them contains the separator $v_i \in \Sigma^+$ of E. So every internal separator of E_j is nonempty. □

3 Exact Reconstruction from Positive Data

The reconstruction operator is Yoshinaka's observed-tuple grammar [14, Section 4.1] with type-guarded substitution links; its soundness and completeness proofs (Sections 3.2 and 3.4) transcribe [14, Lemmas 3–5]. They are given in full because they also cover unbounded rank and because the explicit witness set of Definition 3.10 is what Section 4 measures. Section 3.6 places the bounded-rank case in Yoshinaka's signature framework.

3.1 Set-Driven Reconstruction

Definition 3.1 (Composition witness)**.** Let K be finite and $r \geq 1$. A *composition witness of rank r* in K is a triple $(\vec{z}, \rho, (\vec{u}_1, \ldots, \vec{u}_r))$ with $\vec{z} \in \mathrm{Obs}(K)$, ρ a good template of rank r whose output and child arities are at most f, $\vec{u}_j \in (\Sigma^+)^{d_j}$, and $\vec{z} = \rho(\vec{u}_1, \ldots, \vec{u}_r)$.

Since each $\alpha_i(\vec{u}_1, \ldots, \vec{u}_r) = z_i$ fixes the positions of all child components inside $\vec{z}$, a composition witness is the same as a segmentation of an observed occurrence of $\vec{z}$ into terminal gaps and labelled child intervals. If $E[\vec{z}] \in K$ with $E \in \mathcal{E}_e$, Lemma 2.11 shows that every $\vec{u}_j \in \mathrm{Obs}(K)$. Linearity and nondeletion make the children contribute disjointly to $\vec{z}$, and each contributes at least one letter, so

$$r \leq \textstyle\sum_j \|\vec{u}_j\|_1 \leq \|\vec{z}\|_1 \leq \max_{w \in K} |w| \leq \|K\|_+. \tag{3.1}$$

Definition 3.2 (Capped and uncapped reconstruction operators)**.** Fix f and $h = \widehat{s}$. For $r_{\max} \geq 1$ let $\mathcal{B}_{f,h}^{\leq r_{\max}}(K) = \widehat{G}_{\leq r_{\max}}(K)$ be the grammar with start symbol $\widehat{S}$, a nonterminal $[\vec{x}]$ of arity d for each $\vec{x} \in \mathrm{Obs}_d(K)$, $d \leq f$, and the rules

(Start) $\widehat{S} \to [(w)]$ for $w \in K \cap \Sigma^+$, and $\widehat{S} \to \varepsilon$ if $\varepsilon \in K$;

(Const) $[\vec{x}] \to \vec{x}$ for every observed $\vec{x}$;

(Comp) $[\vec{z}] \to \rho([\vec{u}_1], \ldots, [\vec{u}_r])$ for every composition witness of rank $1 \leq r \leq r_{\max}$;

(Link) $[\vec{x}] \to [\vec{y}]$ for observed $\vec{x}, \vec{y}$ of the same arity d with $h^{(d)}(\vec{x}) = h^{(d)}(\vec{y})$ and $E[\vec{x}], E[\vec{y}] \in K$ for some $E \in \mathcal{E}_d$.

For $K \neq \emptyset$ put $\mathcal{B}_{f,h}(K) := \widehat{G}(K) := \widehat{G}_{\leq \|K\|_+}(K)$; $\mathcal{B}_{f,h}(\emptyset)$ has no rules. Fixed encoding orders make both operators deterministic and set-driven.

By (3.1), $\mathcal{B}_{f,h}(K)$ contains a Comp rule for every composition witness in K. Both operators are sample-consistent, by $\widehat{S} \to [(w)] \to (w)$ and $\widehat{S} \to \varepsilon$. Every $[\vec{x}]$ derives only tuples with nonempty components, since Comp templates are λ-free; so $\widehat{S} \Rightarrow^* \varepsilon$ only by the rule $\widehat{S} \to \varepsilon$. For trivial typing and $\varepsilon \notin K$, Comp/Const, Link and Start are exactly the rules of Types I, II and III of [14, Section 4.1], and $\mathcal{B}_{f,h}$ is the operator of Yoshinaka's unbounded-rank learner $\mathcal{A}(p, *)$ [14, Corollary 2].

3.2 Soundness

Lemma 3.3 (Composition preserves equivalence)**.** *Let $L \subseteq \Sigma^*$, let ρ be a good template of rank r, and let $\vec{x}_j, \vec{u}_j \in (\Sigma^+)^{d_j}$ satisfy $\vec{u}_j \equiv_{L,h} \vec{x}_j$ for every j. Then $\rho(\vec{u}_1, \ldots, \vec{u}_r) \equiv_{L,h} \rho(\vec{x}_1, \ldots, \vec{x}_r)$.*

Proof. Each output component is the concatenation of the terminals of ρ and of child components in a fixed order, and is nonempty by λ-freeness; since h is a monoid morphism, the parent types agree. Fix $E \in \mathcal{E}_e$ and put $\vec{p}_j := \rho(\vec{x}_1, \ldots, \vec{x}_j, \vec{u}_{j+1}, \ldots, \vec{u}_r)$ for $0 \leq j \leq r$. Lemma 2.11 with siblings $\vec{x}_1, \ldots, \vec{x}_{j-1}, \vec{u}_{j+1}, \ldots, \vec{u}_r$ gives $E_j \in \mathcal{E}_{d_j}$ with $E[\vec{p}_{j-1}] = E_j[\vec{u}_j]$ and $E[\vec{p}_j] = E_j[\vec{x}_j]$; as $\mathcal{D}_L(\vec{u}_j) = \mathcal{D}_L(\vec{x}_j)$, $E[\vec{p}_{j-1}] \in L$ iff $E[\vec{p}_j] \in L$. Chaining over j gives $\mathcal{D}_L(\vec{p}_0) = \mathcal{D}_L(\vec{p}_r)$. □

Proposition 3.4 (Soundness of every hypothesis)**.** *Let L be (f, s)-tuple-substitutable and $K \subseteq L$ finite. For every $r_{\max} \geq 1$, if $[\vec{x}] \Rightarrow^* \vec{u}$ in $\widehat{G}_{\leq r_{\max}}(K)$, then $\vec{u} \equiv_{L,h} \vec{x}$. Consequently $L(\mathcal{B}_{f,h}^{\leq r_{\max}}(K)) \subseteq L$ and $L(\mathcal{B}_{f,h}(K)) \subseteq L$.*

Proof. This is [14, Lemmas 3–4] with typed links. Induct on the derivation, according to its first rule. Const: $\vec{u} = \vec{x}$. Link $[\vec{x}] \to [\vec{y}]$: $\vec{x}, \vec{y}$ have equal types and share a context in $K \subseteq L$, so substitutability gives $\vec{x} \equiv_{L,h} \vec{y}$, and $\vec{u} \equiv_{L,h} \vec{y}$ by induction. Comp $[\vec{z}] \to \rho([\vec{u}_1], \ldots, [\vec{u}_r])$ with $[\vec{u}_j] \Rightarrow^* \vec{v}_j$: by induction $\vec{v}_j \equiv_{L,h} \vec{u}_j$, and Lemma 3.3 gives $\rho(\vec{v}_1, \ldots, \vec{v}_r) \equiv_{L,h} \vec{z}$. A nonempty word w is derived from $\widehat{S}$ via some $[(w_0)]$ with $w_0 \in K$; then $(w) \equiv_{L,h} (w_0)$ and $\square_1 \in \mathcal{D}_L((w_0))$, so $w \in L$; and ε is derived only if $\varepsilon \in K \subseteq L$. The uncapped operator is the capped one with $r_{\max} = \|K\|_+$. □

3.3 Yield-Typed Refinement and Witnesses

Definition 3.5 (Yield-type refinement)**.** Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a finite nondeleting MCFG and $h : \Sigma^* \to M$ a finite-monoid morphism. For a template ρ and $\mathbf{t}_j \in M^{d_j}$, let $\mathrm{out}_\rho(\mathbf{t}_1, \ldots, \mathbf{t}_r) \in M^{d_0}$ have ith component the image of α_i under the monoid morphism $\Gamma_\rho^* \to M$ extending $a \mapsto h(a)$ and $\xi_j^\kappa \mapsto (\mathbf{t}_j)_\kappa$. The *full yield-type refinement* G^h has a fresh start symbol $\widetilde{S}$, nonterminals $A_{\mathbf{t}}$ for $A \in V$

and $\mathbf{t} \in M^{\mathrm{fan}(A)}$, start rules $\widetilde{S} \to S_{(\eta)}$ for $\eta \in M$, and for every rule $A \to \rho(B_1, \dots, B_r)$ (including $r = 0$) and all $\mathbf{t}_j \in M^{\mathrm{fan}(B_j)}$ the rule $A_{\mathrm{out}_\rho(\mathbf{t}_1,\dots,\mathbf{t}_r)} \to \rho(B_{1,\mathbf{t}_1}, \dots, B_{r,\mathbf{t}_r})$. The *trimmed refinement* $\widetilde{G}_0$ keeps exactly the typed nonterminals and rules occurring in some successful derivation from $\widetilde{S}$. Unless G^h is named, *typed nonterminal* and *typed rule* refer to the non-start nonterminals and rules of $\widetilde{G}_0$, and $L_X := \{\vec{u} : X \Rightarrow^*_{\widetilde{G}_0} \vec{u}\}$.

Typed rules have the templates of their sources, so $\widetilde{G}_0$ is good whenever G is; each L_X is nonempty, with nonempty components if G is λ-free.

Proposition 3.6 (Yield-type invariants)**.** *(i) If $A_{\mathbf{t}} \Rightarrow^* \vec{u}$ in G^h, then $h^{(\mathrm{fan}(A))}(\vec{u}) = \mathbf{t}$, and erasing indices gives a G-derivation of $\vec{u}$.*

(ii) Every terminal derivation $A \Rightarrow^*_G \vec{u}$ lifts uniquely to $A_{h^{(\mathrm{fan}(A))}(\vec{u})} \Rightarrow^*_{G^h} \vec{u}$; a successful derivation of w lifts uniquely through $\widetilde{S} \to S_{(h(w))}$.

(iii) $L(\widetilde{G}_0) = L(G^h) = L(G)$.

Proof. At every rule application $h^{(\mathrm{fan}(A))}(\rho(\vec{u}_1, \dots, \vec{u}_r)) = \mathrm{out}_\rho(h^{(d_1)}(\vec{u}_1), \dots, h^{(d_r)}(\vec{u}_r))$, because h is a monoid morphism. Induction on derivation height gives (i) and the unique lifting (ii). Erasing indices gives $L(G^h) \subseteq L(G)$, (ii) the converse, and trimming keeps every rule of every successful derivation. □

Lemma 3.7 (Monoid implementations)**.** *Let $h_0 : \Sigma^* \to M_0$ be a finite-monoid morphism, $S_0 := h_0(\Sigma^+)$ and $s_0 := h_0|_{\Sigma^+}$.*

(i) *For $1 \le d \le f$ and $\vec{x}, \vec{y} \in (\Sigma^+)^d$, the equalities $h_0^{(d)}(\vec{x}) = h_0^{(d)}(\vec{y})$, $s_0^{(d)}(\vec{x}) = s_0^{(d)}(\vec{y})$ and $\widehat{s_0}^{(d)}(\vec{x}) = \widehat{s_0}^{(d)}(\vec{y})$ are equivalent. Hence $\mathcal{C}^{\mathrm{mcf}}_{f,h_0} = \mathcal{C}^{\mathrm{mcf}}_{f,s_0}$, and the reconstruction operators built with h_0 and with $\widehat{s_0}$ return the same grammar on every sample.*

(ii) *For every good MCFG G, the trimmed refinements of G under h_0 and under $\widehat{s_0}$ coincide (the untrimmed ones need not); hence the same canonical tuples, contexts and witness words may be chosen for both.*

Proof. (i) The three maps agree on Σ^+ as maps into S_0, and the class and the Link guard compare types of nonempty components only. (ii) S_0 is a subsemigroup of M_0 and of $S_0^\sharp$ with the same multiplication. By λ-freeness and Proposition 3.6(i), every typed nonterminal occurring in a successful derivation under either morphism is some $A_{\mathbf{t}}$ with $\mathbf{t} \in S_0^{\mathrm{fan}(A)}$, and the start links used are $\widetilde{S} \to S_{(\eta)}$ with $\eta \in S_0$. For child indices in S_0, every component α_i of a good template is a nonempty word whose letters and variables are evaluated in S_0 identically under both morphisms, so both give the same out_ρ. Thus the two full refinements contain the same rules among S_0-indexed symbols, which carry all successful derivations. □

Lemma 3.8 (Canonical contexts for typed nonterminals)**.** *Let G be good. Every typed nonterminal X of arity d has a context $\chi(X) \in \mathcal{E}_d$ with $\chi(X)[\vec{u}] \in L(G)$ for all $\vec{u} \in L_X$; if $\widetilde{S} \to X$ is in $\widetilde{G}_0$, one may take $\chi(X) = \Box_1$.*

Proof. Take a successful typed derivation containing X, and follow the path from the root to an occurrence of X, starting with $\Box_1$. At each rule on the path apply Lemma 2.11 with the (nonempty) yields of the sibling subderivations fixed; this gives $\chi(X) \in \mathcal{E}_d$. The rules on the path depend only on the label X, so replacing the subtree of X by any X-derivation yields a successful derivation of $\chi(X)[\vec{u}]$. If $\widetilde{S} \to X$ is a rule, use the path of length zero. □

Lemma 3.9 (Typed nonterminals are distributionally homogeneous)**.** *Let G be good with $L := L(G) \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$. For every typed nonterminal X of arity d, L_X lies in a single $\equiv^d_{L,h}$-class, denoted $[X]$.*

Proof. For $\vec{u}, \vec{v} \in L_X$, Proposition 3.6(i) gives equal types and $\chi(X)$ accepts both; substitutability gives $\mathcal{D}_L(\vec{u}) = \mathcal{D}_L(\vec{v})$. □

This parallels, but is weaker than, the homogeneity of congruential MCFGs [17], whose contexts may have empty separators.

Definition 3.10 (Canonical tuples, contexts and the witness set)**.** Let G be good. For every typed nonterminal X fix a *canonical tuple* $\omega(X) \in L_X$ and a *canonical context* $\chi(X)$ as in Lemma 3.8, with $\chi(X) = \square_1$ whenever $\widetilde{S} \to X$ is in $\widetilde{G}_0$. Let $\mathsf{W}(\widetilde{G}_0)$ consist of the words

$$\chi(X)[\omega(X)] \quad (X \text{ typed}), \qquad \chi(X)[\rho(\omega(Y_1), \ldots, \omega(Y_q))] \quad (X \to \rho(Y_1, \ldots, Y_q) \text{ typed}, \ q \geq 0).$$

Since $\omega(X)$ and $\rho(\omega(Y_1), \ldots, \omega(Y_q))$ belong to L_X, $\mathsf{W}(\widetilde{G}_0)$ is a finite subset of $L(G)$, nonempty if $L(G) \neq \emptyset$. Components of canonical tuples are nonempty, so $\|\omega(X)\|_1 \geq \mathrm{fan}(X)$. Canonical objects are target witnesses, not inputs of the operator; qualitatively any choice works, while Section 4 chooses short ones.

Proposition 3.11 (Cardinality of the witness set)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be good of fan-out $\leq f$ and rule rank $\leq r_{\max}$, and let n_{typ} and n_{rule} be the numbers of typed nonterminals and typed rules of $\widetilde{G}_0$. Then*

$$|\mathsf{W}(\widetilde{G}_0)| \leq n_{\mathrm{typ}} + n_{\mathrm{rule}} \leq |V||M|^f + |P||M|^{f r_{\max}}.$$

Proof. One word is selected per typed nonterminal and per typed rule. A typed nonterminal is some $A_{\mathbf{t}}$ with $\mathbf{t} \in M^{\mathrm{fan}(A)}$, and a typed rule is determined by its source rule and the child indices $\mathbf{t}_j \in M^{\mathrm{fan}(B_j)}$. □

3.4 Completeness and Exact Reconstruction

The completeness argument [14, Lemma 5] needs only that the sample exposes, for every typed rule, one same-type link and one decomposition. We state it for arbitrary typed representatives, so that it also applies to the relaxed witnesses of Section 4. No assumption that $\iota(X) \in L_X$ is made: the two required witnessed sentence words are instead explicit hypotheses.

Proposition 3.12 (Completeness from a witness assignment)**.** *Let G be a finite good MCFG of fan-out $\leq f$, let r_G be the maximum rank of a typed rule of $\widetilde{G}_0$ ($r_G := 0$ if none), and assign to every typed nonterminal $X = A_{\mathbf{t}}$ a tuple $\iota(X) \in (\Sigma^+)^{\mathrm{fan}(A)}$ with $h^{(\mathrm{fan}(A))}(\iota(X)) = \mathbf{t}$ and a context $\chi(X) \in \mathcal{E}_{\mathrm{fan}(A)}$, where $\chi(X) = \square_1$ whenever $\widetilde{S} \to X$ is in $\widetilde{G}_0$. If a finite sample K contains $\chi(X)[\iota(X)]$ for every X and $\chi(X)[\rho(\iota(Y_1), \ldots, \iota(Y_q))]$ for every typed rule $X \to \rho(Y_1, \ldots, Y_q)$, $q \geq 0$, then*

$$L(G) \subseteq L(\mathcal{B}^{\leq r_{\max}}_{f,h}(K)) \quad (r_{\max} \geq \max\{1, r_G\}), \qquad L(G) \subseteq L(\mathcal{B}_{f,h}(K)).$$

Substitutability is not assumed. The canonical choice $\iota = \omega$ gives the hypothesis $\mathsf{W}(\widetilde{G}_0) \subseteq K$.

Proof. We show that $X \Rightarrow^*_{\widetilde{G}_0} \vec{u}$ implies $[\iota(X)] \Rightarrow^* \vec{u}$ in both grammars, by induction on the derivation. Each $\iota(X)$ is observed, since $\chi(X)[\iota(X)] \in K$. Let the derivation start with $X \to \rho(Y_1, \ldots, Y_q)$, followed by $Y_j \Rightarrow^* \vec{u}_j$, and put $\vec{z} := \rho(\iota(Y_1), \ldots, \iota(Y_q))$. Then $\vec{z}$ is observed, since $\chi(X)[\vec{z}] \in K$, and $h(\vec{z}) = \mathrm{out}_\rho(\text{indices of the } Y_j)$ is the index of X, which is $h(\iota(X))$; as $\chi(X)$ accepts both in K, the rule $[\iota(X)] \to [\vec{z}]$ is a Link. If $q = 0$, Const gives $[\vec{z}] \to \vec{z} = \vec{u}$. If $q \geq 1$, $(\vec{z}, \rho, (\iota(Y_j))_j)$ is a composition

witness of rank $q \le r_G \le r_{\max}$, and $q \le \|K\|_+$ by (3.1), so Comp gives $[\vec{z}] \to \rho([\iota(Y_1)], \dots, [\iota(Y_q)])$ in both operators; the induction hypothesis completes $[\iota(X)] \Rightarrow^* \rho(\vec{u}_1, \dots, \vec{u}_q)$. Finally a successful derivation begins with $\widetilde{S} \to X$, where $\chi(X) = \square_1$, so $\iota(X) = (w_0)$ with $w_0 \in K \cap \Sigma^+$ and $\widehat{S} \to [\iota(X)]$ is a Start rule; by Proposition 3.6(iii) this covers $L(G)$. □

Theorem 3.13 (Exact reconstruction for a fixed typing)**.** *Fix $f \ge 1$ and $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$, and put $h = \widehat{s}$. Let $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$. If $L \cap \Sigma^+ = \emptyset$ put $K^\star_L := L$ and $r_G := 0$. Otherwise let G be a good presentation of $L \cap \Sigma^+$ (Proposition 2.6), and put $K^\star_L := \mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$. Then $K^\star_L \subseteq L$ is finite, and every finite K with $K^\star_L \subseteq K \subseteq L$ satisfies*

$$L(\mathcal{B}_{f,s}(K)) = L, \qquad L(\mathcal{B}^{\le r_{\max}}_{f,s}(K)) = L \quad \text{for every } r_{\max} \ge \max\{1, r_G\}.$$

If $L \in \mathsf{MCFL}(f, r)$, G can be chosen of rank $\le r$, so $r_{\max} = \max\{1, r\}$ suffices (also when $r = 0$).

Proof. If $L \cap \Sigma^+ = \emptyset$ then $K = L \in \{\emptyset, \{\varepsilon\}\}$ and the claim is immediate from Definition 3.2. Otherwise $K^\star_L \subseteq L$ is finite by the remark after Definition 3.10. Proposition 3.12 with $\iota = \omega$ (admissible by Proposition 3.6(i)) gives $L \cap \Sigma^+ \subseteq L(\mathcal{B}(K))$, and Proposition 3.4 gives $L(\mathcal{B}(K)) \subseteq L$, for both operators. The hypothesis contains ε iff $\varepsilon \in K$, iff $\varepsilon \in L$, because $K^\star_L \subseteq K \subseteq L$. Rank-preserving normalization gives the last sentence. □

3.5 Effective Reconstruction and Conservative Learning

Theorem 3.14 (Polynomial construction under a rule-rank cap)**.** *Fix $f, r_{\max} \ge 1$ and s, and implement type tests by $h = \widehat{s}$. The encoding of $\widehat{G}_{\le r_{\max}}(K)$ is computable in time $(1 + \|K\|_+)^{O(f r_{\max})}$; its fan-out is at most f and its rank at most $r_{\max}$.*

Proof. Put $n_K = \max\{1, \|K\|_+\}$. An occurrence is a sample word with at most $2f$ cut positions, so there are $n_K^{O(f)}$ occurrences, typed by fixed-monoid products; grouping occurrences by context and type lists the Link rules. A composition witness of rank $r \le r_{\max}$ is determined by a parent occurrence and at most $2f r_{\max}$ further cut positions inside it, with a labelling whose number depends only on $f, r_{\max}$; goodness is checked directly. This gives $n_K^{O(f r_{\max})}$ candidate Comp rules; the remaining rules have rank at most one, and deduplication leaves an encoding of the same size. □

Definition 3.15 (Conservative sequential learner)**.** The learner $\mathcal{A}_{f,h}$ (respectively $\mathcal{A}^{\le r_{\max}}_{f,h}$) starts with H_0, keeps its hypothesis on $\perp$ and on every datum the hypothesis generates, and otherwise replaces it by $\mathcal{B}_{f,h}(K)$ (respectively $\mathcal{B}^{\le r_{\max}}_{f,h}(K)$), where K is the content read so far. This is the policy of [14, Algorithm 1].

Corollary 3.16 (Gold identification)**.** *Fix $f \ge 1$ and s.*

(i) *$\mathcal{A}_{f,s}$ TxtEx-identifies $\mathcal{C}^{\mathrm{mcf}}_{f,s}$.*

(ii) *For fixed $r_{\max}$, $\mathcal{A}^{\le r_{\max}}_{f,s}$ TxtEx-identifies $\mathcal{C}^{\mathrm{mcf}}_{f,s} \cap \mathsf{MCFL}(f, r_{\max})$, in time per datum polynomial in the size of the content read so far.*

Proof. The learners are computable: hypotheses are computable and MCFG membership is decidable [30]. Let L be in the class and fix a text; let $K^\star_L$ be as in Theorem 3.13, with $r_{\max} \ge r_G$ in case (ii), and let n_0 be a stage whose content contains $K^\star_L$. Every hypothesis H satisfies $L(H) \subseteq L$, by Proposition 3.4 or because $H = H_0$. A rebuild at a stage $\ge n_0$ yields an exact hypothesis, which is never changed afterwards since all later data lie in L. If no rebuild occurs after n_0, let H be the

hypothesis current at n_0. If $H = H_0$, no positive datum was ever read, so $L = \emptyset = L(H)$. Otherwise H was built from the content of some stage m; it generates that content by sample-consistency and every later datum by conservativeness, hence all of L, so $L(H) = L$. In both cases the outputs stabilize on a correct grammar. For (ii), Theorem 3.14 bounds rebuilding, and membership of a datum w in a hypothesis $\widehat{G}$ of fan-out f and rank $r_{\max}$ takes $O(\|\widehat{G}\|^2 |w|^{f(r_{\max}+1)})$ time [14, Proposition 4]. □

Every CFG has a rank-two presentation, so $\mathcal{A}^{\leq 2}_{1,h}$ covers $\mathcal{C}^{\mathrm{mcf}}_{1,h}$; this does not imply polynomial characteristic-data length at rank two (Section 4).

3.6 Specialization of Yoshinaka's 2015 Signature

We recall from [15, Sections 4 and 6.1] the notions used. A *signature* consists of pairwise disjoint sorts (sets of objects), a distinguished sort O_*, and typed function names; a grammar over it has rules $X_0 \leftarrow g\langle X_1, \ldots, X_n\rangle$ respecting sorts and initial symbols of sort O_*. $\mathcal{S}_O$ and $\mathcal{C}_O$ are the substructures and contexts (functions $O \to O_*$) of sort O realized in some grammar over the signature, and a language is *substitutable* if $c_1 \odot s_1, c_1 \odot s_2, c_2 \odot s_1 \in L$ imply $c_2 \odot s_2 \in L$ for all sorts O, $c_i \in \mathcal{C}_O$, $s_i \in \mathcal{S}_O$.

Fix $f, r \geq 1$ and $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$, with $h = \widehat{s}$. The signature $\mathfrak{S}_{f,r,s}$ has the sort O_*, a tagged copy $\{\overline{w} : w \in \Sigma^*\}$ of Σ^*, and the tagged sorts $T_{d,\boldsymbol{\mu}} = \{\vec{w} \in (\Sigma^+)^d : s^{(d)}(\vec{w}) = \boldsymbol{\mu}\}$ for $1 \leq d \leq f$ and $\boldsymbol{\mu} \in \mathsf{S}^d$. Its functions are: every good template ρ of rank $q \leq r$ with arities at most f, typed as $T_{d_1,\boldsymbol{\mu}_1} \times \cdots \times T_{d_q,\boldsymbol{\mu}_q} \to T_{d_0,\mathrm{out}_\rho(\boldsymbol{\mu}_1,\ldots,\boldsymbol{\mu}_q)}$ for all $\boldsymbol{\mu}_j$ (the index lies in S^{d_0} by λ-freeness); promotions $\pi_\mu : T_{1,\mu} \to O_*$, $(w) \mapsto \overline{w}$; and a constant $\overline{\varepsilon}$ of sort O_*. Forgetting types, this is a finite sort refinement in the sense of [15, Section 8] of the signature of good MCFG operations on nonempty tuples, a nonempty-component variant of [15, Example 3]; at fan-out one the prefix–suffix case corresponds to [15, Example 4].

Proposition 3.17 (Bounded-rank coverage by Yoshinaka's 2015 framework)**.** (i) *$\mathcal{S}_{T_{d,\mu}} = T_{d,\mu}$, $\mathcal{C}_{T_{d,\boldsymbol{\mu}}} = \{\vec{x} \mapsto \overline{E[\vec{x}]} : E \in \mathcal{E}_d\}$ as sets of functions on $T_{d,\boldsymbol{\mu}}$, and $\mathcal{C}_{O_*}$ contains only the identity.*

(ii) $\mathsf{Subst}(\mathfrak{S}_{f,r,s}) = \mathcal{C}^{\mathrm{mcf}}_{f,s} \cap \mathsf{MCFL}(f,r)$.

(iii) *For fixed f, r, s, $\mathfrak{S}_{f,r,s}$ satisfies Assumptions 1–3 of [15]. Hence, by [15, Theorem 1] (proved there for* SUBST$_{\mathrm{D}}$ *via its Lemmas 2–3), $\mathcal{C}^{\mathrm{mcf}}_{f,s} \cap \mathsf{MCFL}(f,r)$ is identifiable in the limit from positive data with polynomial-time updates, and for a target grammar without useless symbols, with N nonterminals and P rules,* SUBST$_{\mathrm{D}}$ *has a characteristic set of cardinality at most $|N| + |P|$ (the bound stated after [15, Lemma 3]).*

Proof. (i) Constants (good rank-zero templates, i.e. tuples of nonempty words) give $\mathcal{S}_T = T$. No function takes an argument of sort O_* and only promotions and $\overline{\varepsilon}$ have codomain O_*; so $\mathcal{C}_{O_*}$ is the identity, and the context of a non-start nonterminal is obtained from the argument context $\square_1$ of a promotion by descending through good templates whose sibling arguments are nonempty tuples. By Lemma 2.11 each step yields an admissible context, so $\mathcal{C}_T \subseteq \{E[\cdot] : E \in \mathcal{E}_d\}$. Conversely $E = v_0 \square_1 v_1 \cdots \square_d v_d \in \mathcal{E}_d$ is the context of Z in the grammar $X_* \leftarrow \pi_\nu\langle Y\rangle$, $Y \leftarrow (v_0 \xi^1_1 v_1 \cdots \xi^d_1 v_d)\langle Z\rangle$, $Z \leftarrow \vec{x}_0$ for any $\vec{x}_0 \in T_{d,\boldsymbol{\mu}}$; the unary template is good because $v_1, \ldots, v_{d-1} \neq \varepsilon$ and its output is nonempty, and ν is its output type.

(ii) By (i), Yoshinaka's condition for the sort $T_{d,\boldsymbol{\mu}}$ says: for $E_1, E_2 \in \mathcal{E}_d$ and $\vec{x}, \vec{y}$ of type $\boldsymbol{\mu}$, $E_1[\vec{x}], E_1[\vec{y}], E_2[\vec{x}] \in L$ imply $E_2[\vec{y}] \in L$; for O_* it is vacuous. Over all $d \leq f$ and $\boldsymbol{\mu}$ this is (f,s)-tuple substitutability. It remains to see that $\mathfrak{S}_{f,r,s}$-grammars generate exactly $\mathsf{MCFL}(f,r)$. Erasing tags turns such a grammar into an MCFG of fan-out $\leq f$ and rank $\leq r$ (promotions become identity rules into a fresh start symbol, $\overline{\varepsilon}$ a constant rule). Conversely, for $L \in \mathsf{MCFL}(f,r)$ with $L \cap \Sigma^+ \neq \emptyset$,

take a good presentation G of $L \cap \Sigma^+$ of rank $\leq r$ (Proposition 2.6). In $\widetilde{G}_0$ every non-start index lies in $\mathsf{S}^{\mathrm{fan}(A)}$ and every typed rule is an instance of a function of $\mathfrak{S}_{f,r,s}$; replace each start rule $\widetilde{S} \to S_{(\eta)}$ by a promotion π_η and add $\overline{\varepsilon}$ if $\varepsilon \in L$. The result generates L by Proposition 3.6(iii). The case $L \subseteq \{\varepsilon\}$ is trivial.

(iii) Assumption 2 holds since arities are at most r. For Assumption 1, sort membership, function evaluation, recognizing admissible contexts (by (i)), plugging, and membership in the language of a grammar over $\mathfrak{S}_{f,r,s}$ are polynomial; the last is bounded fan-out and rank MCFG membership after erasing tags [14, Proposition 4]. For Assumption 3, let $d \in \Sigma^*$. By (i) the substructures of non-start sorts occurring in d are the tuples of occurrences of arity $\leq f$ in d, typed by s, and their contexts are the corresponding admissible contexts; at O_* they are $\overline{d}$ and the identity. A typed template ρ with $c \odot \rho(\vec{u}_1, \ldots, \vec{u}_q) = \overline{d}$ yields an occurrence of $\vec{z} = \rho(\vec{u}_1, \ldots, \vec{u}_q)$ in d, i.e. a composition witness on it (a constant if $q = 0$); promotions and $\overline{\varepsilon}$ occur only for $d \neq \varepsilon$ and $d = \varepsilon$, respectively. Crucially, a function *name* includes its output sort and the ordered list of input sorts: for every candidate factorization the componentwise s-types of the observed child tuples determine these sorts, and the output sort is computed by out_ρ. Thus the enumeration records each typed instance, not only the untyped template, including the well-typed unary identity instances. As in Theorem 3.14, all these sets are computable in time $(|d| + 1)^{O(f(r+1))}$. □

Lemma 3.18 (Substructure-indexed and guarded reconstruction)**.** *Let* $\mathrm{SUBST_P}$ *be Yoshinaka's substructure-indexed construction [15, Section 6.1] for* $\mathfrak{S}_{f,r,s}$ *with the sets of the preceding proof. For every finite* $K \subseteq \Sigma^*$, $L(\mathrm{SUBST_P}(K)) = L(\mathcal{B}^{\leq r}_{f,s}(K))$.

Proof. $\mathrm{SUBST_P}(K)$ has a nonterminal $[\![x]\!]$ for each substructure x occurring in K, initial symbols $[\![\overline{w}]\!]$ for $w \in K$, and a rule $[\![x_0]\!] \leftarrow g\langle [\![x_1]\!], \ldots, [\![x_n]\!] \rangle$ for each function g occurring in K and occurring substructures x_i of matching sorts whenever some context c occurring in K has $c \odot x_0, c \odot g(x_1, \ldots, x_n) \in K$. On non-start sorts identify $[\![\vec{x}]\!]$ with $[\vec{x}]$; these range over $\mathrm{Obs}(K)$.

Such a rule with $g = \rho$ of rank n is simulated in $\mathcal{B}^{\leq r}_{f,s}(K)$: put $\vec{z} := \rho(\vec{x}_1, \ldots, \vec{x}_n)$; by (i), $c = E[\cdot]$ with $E \in \mathcal{E}_d$ and $E[\vec{x}_0], E[\vec{z}] \in K$, so $\vec{z}$ is observed and has the sort, hence the type, of $\vec{x}_0$; thus $[\vec{x}_0] \to [\vec{z}]$ is a Link, followed by Const $[\vec{z}] \to \vec{z}$ if $n = 0$ or by the Comp rule of the composition witness $(\vec{z}, \rho, (\vec{x}_j)_j)$ if $n \geq 1$. Conversely, each non-start rule of $\mathcal{B}^{\leq r}_{f,s}(K)$ is a rule of $\mathrm{SUBST_P}(K)$: Const $[\vec{x}] \to \vec{x}$ and Comp $[\vec{z}] \to \rho(\ldots)$ use $x_0 = \vec{x}$, resp. $x_0 = \vec{z}$, and the context of an occurrence of x_0; Link $[\vec{x}] \to [\vec{y}]$ uses the identity template of the common sort, which occurs in K, and the shared context. For the Link case, the unary identity has a genuinely observed function instance: if $E[\vec{y}] \in K$, then $E[\mathrm{id}(\vec{y})] \in K$; hence its typed function name belongs to $\mathrm{FUN}(K)$. Likewise, each Comp uses its witnessed observed parent and child factorization, and each Const its rank-zero instance. By induction on derivations, $[\![\vec{x}]\!]$ and $[\vec{x}]$ derive the same tuples. At O_* the only context is the identity and only promotions and $\overline{\varepsilon}$ have codomain O_*; so for $K \neq \emptyset$ the rules for initial symbols are $[\![\overline{w}]\!] \leftarrow \pi\langle [\![(w')]\!] \rangle$ $(w, w' \in K,\ w' \neq \varepsilon)$ and $[\![\overline{w}]\!] \leftarrow \overline{\varepsilon}$ $(w \in K, \varepsilon \in K)$, and both grammars generate $\bigcup_{w' \in K \cap \Sigma^+} L([(w')]) \cup (\{\varepsilon\} \cap K)$; for $K = \emptyset$ both languages are empty. □

Remark 3.19 (What is and is not inherited). (a) Yoshinaka proves [15, Theorem 1] for $\mathrm{SUBST_D}$ and states without separate proof that his Algorithm 1 also works with $\mathrm{SUBST_P}$ [15, Sections 6.1 and 7]. For $\mathfrak{S}_{f,r,s}$, Lemma 3.18 together with Proposition 3.4 and Theorem 3.13 supplies that verification; Lemma 3.18 asserts equality of generated languages, not of grammars. (b) Cardinality bounds, whether $|N| + |P|$ or Proposition 3.11, do not bound total length; this is the subject of Section 4. (c) Assumption 2 bounds arity in each signature, so Proposition 3.17 covers $\mathsf{MCFL}(f, r)$ only for fixed r; unbounded rank is covered by Theorem 3.13, extending [14, Corollary 2] to typed links.

4 Characteristic-Data Bounds

Under a rule-rank cap, Proposition 3.11 already gives polynomially many characteristic words; this section controls their total length.

Condition	Characteristic-data control	Source
arbitrary h, bounded rank	polynomial in $\|G\|$ and refined thickness $\theta_{\widetilde{G}_0}$	Lemma 4.3
rule rank one	polynomial in $\|G\|$ alone	Theorem 4.5
$h \preceq h_{\Sigma}^{k,\ell}$, bounded rank	$O(\|G\|^2\bar{\theta}_G)$	Theorem 4.9
window plus any finite group	$O(\|G\|^3\bar{\theta}_G)$	Theorem C.7

The first row is the unconditional baseline for general h. Full yield-type refinement can make its thickness exponentially larger than the original presentation parameters (Proposition 4.4), so that route alone gives no original-thickness bound. Rank one removes the thickness dependence, whereas prefix–suffix-determined typing gives the main higher-rank transfer. The finite-abelian-group refinement has its direct zero-sum proof in Appendix B; the nonabelian extension is established by a separate order-homogenization proof in Appendix C. The companion CFG result [25] separately excludes a universal original-thickness polynomial for arbitrary finite typing at fan-out one; the semilattice typing of that example is not determined by any fixed prefix–suffix window. Polynomial update time in the observed sample is independent of these data-length refinements (Theorem 3.14 and Corollary 3.16).

4.1 Thickness and reusable canonical contexts

Rule thickness and Yoshinaka's efficiency criterion. For the quantitative results, presentations are reduced and *good*, as in Yoshinaka's fixed-dimension, fixed-rank good form [14, Section 2.3, after Example 2]. For a rule R of G let $\theta_G(R)$ be the minimum length of a successful sentence derivation using R, put $\theta_G := \max_{R\in P} \theta_G(R)$, and define $t_G := \max_A \min\{\|\vec{w}\|_1 : \vec{w} \in L_A(G)\}$. Thus θ_G is rule thickness and t_G nonterminal thickness. Yoshinaka writes these quantities as τ_G and t_G, respectively, with $\tau_G \le \|G\| t_G$ in his notation [14, Section 4.3].

Lemma 4.1 (Polynomial equivalence of the two thickness parameters)**.** *For every reduced good presentation G, $t_G \le \theta_G$ and $\theta_G \le 2\|G\| \max\{1, t_G\}$.*

Proof. For $t_G \le \theta_G$, choose any rule with left-hand side A and place it in a successful derivation. Nondeletion preserves the tuple produced at that occurrence of A, so its total length is bounded by the final sentence. Minimize for A and maximize over A.

For the converse, fix a rule R with left-hand side A and follow a shortest dependency path from the start symbol to A. The path repeats no nonterminal. Keep the path child open, complete each sibling by a shortest terminal tuple, then apply R and complete its children likewise. Across the distinct rules on this construction, both explicit terminal material and child-variable occurrences are bounded by $\|G\|$, giving a successful sentence of length at most $\|G\| + \|G\| t_G \le 2\|G\| \max\{1, t_G\}$. □

Hence the two thickness parameters are polynomially equivalent on reduced good presentations. Yoshinaka's efficiency criterion asks for polynomial hypothesis construction, characteristic-set cardinality, and total positive data in target size and rule thickness [14, Definition 1]; see also [10]. Here characteristic-data bounds concern the capped set-driven reconstruction, while TxtEx stabilization is supplied by the conservative learner. Proposition A.1 transfers the size–thickness scale for reduced λ-free inputs; genuine empty components may instead cause the exponential growth of Proposition A.2.

Lemma 4.2 (Reusable canonical-context bound)**.** *Let $\mathcal{G}_{\mathrm{t}}$ be a trimmed good typed MCFG with n_{typ} non-start typed nonterminals and maximum non-start rule rank at most $r_0 \geq 1$. Suppose that every non-start typed nonterminal Y has a canonical tuple $\omega(Y) \in L_Y(\mathcal{G}_{\mathrm{t}})$ with $\|\omega(Y)\|_1 \leq b_{\mathrm{can}}$, and that every non-start typed rule writes at most b_{rule} terminal symbols explicitly. Then the canonical contexts may be chosen so that*

$$|\chi(X)|_{\mathrm{fix}} \leq n_{\mathrm{typ}}(b_{\mathrm{rule}} + (r_0 - 1)b_{\mathrm{can}})$$

for every non-start typed nonterminal X. With these contexts and canonical tuples, every selected witness word has length at most

$$n_{\mathrm{typ}}(b_{\mathrm{rule}} + (r_0 - 1)b_{\mathrm{can}}) + b_{\mathrm{rule}} + r_0 b_{\mathrm{can}}.$$

Proof. If $\widetilde{S} \to X$ is present take $\chi(X) = \Box_1$; otherwise follow a shortest typed-nonterminal dependency path to X. It has at most n_{typ} non-start steps. Keeping the path child open and filling each sibling canonically costs at most $b_{\mathrm{rule}} + (r_0 - 1)b_{\mathrm{can}}$ terminals per step. Goodness preserves the selected components, their order, and nonempty separators, giving the context bound. Adding either one canonical tuple or one rule template with at most r_0 child tuples gives the displayed witness bound. □

4.2 Refined-Thickness Baseline

For a reduced good presentation G of an ε-free target, let $\widetilde{G}_0$ be its trimmed full yield-type refinement and put $\bar{\theta}_{\mathrm{typ}} := \max\{1, \theta_{\widetilde{G}_0}\}$, where $\theta_{\widetilde{G}_0}$ is rule thickness computed in the refined grammar. No comparison with the original untyped thickness is assumed.

Lemma 4.3 (Refined-thickness baseline)**.** *Fix $f \geq 1$, $r_{\max} \geq 1$, a finite alphabet Σ, and a finite typing $h : \Sigma^* \to M$. Let G be a reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ presentation of an ε-free language in $\mathcal{C}^{\mathrm{mcf}}_{f,h}$, and let $\widetilde{G}_0$ be its trimmed yield-type refinement. Then the target witness set has cardinality polynomial in $\|G\|$ and total positive length polynomial in $\|G\|$ and $\bar{\theta}_{\mathrm{typ}} := \max\{1, \theta_{\widetilde{G}_0}\}$.*

Proof. The trimmed refinement $\widetilde{G}_0$ remains reduced and good. A minimum-yield successful derivation using a typed rule with left-hand side X contains an X-tuple of total length at most $\bar{\theta}_{\mathrm{typ}}$, so use such tuples canonically. If $\widetilde{S} \to X$ exists take $\chi(X) = \Box_1$; otherwise remove the chosen X-subtree from the same derivation to obtain a canonical context with fixed material at most $\bar{\theta}_{\mathrm{typ}}$. Each typed rule writes at most that many terminals and has at most $r_{\max}$ canonical children, so every selected witness has length at most $(r_{\max} + 2)\bar{\theta}_{\mathrm{typ}}$. Therefore

$$|\mathsf{W}(\widetilde{G}_0)| \leq |V||M|^f + |P||M|^{f r_{\max}}, \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ \leq (|V||M|^f + |P||M|^{f r_{\max}})(1 + (r_{\max} + 2)\bar{\theta}_{\mathrm{typ}}),$$

which is polynomial for fixed $f, r_{\max}, h$. □

For trivial typing, no nonterminal is split by yield type; consequently the original thickness scale is retained. This recovers Yoshinaka's untyped characteristic-data estimate [14, Lemma 7]. Without this uniqueness, the refinement may have much larger thickness, as the next proposition demonstrates.

Proposition 4.4 (Exponential blow-up of full yield-type refinement)**.** *There are a fixed two-element typing h_{c}, reduced good fan-out-one rank-two grammars G_n of size $O(n)$, and a constant bound $\theta_{G_n} \leq 2$, such that $L(G_n) = \{\mathsf{a}, \mathsf{c}\}^+$ and the trimmed full yield-type refinement $\widetilde{G}_{n,0}$ satisfies $\theta_{\widetilde{G}_{n,0}} \geq 2^n$.*

Proof. Let $M_{\mathtt{c}} = \{1, \zeta\}$ with $\zeta^2 = \zeta$, $h_{\mathtt{c}}(\mathtt{a}) = 1$, and $h_{\mathtt{c}}(\mathtt{c}) = \zeta$. For $n \geq 1$, view

$$S \to SS \mid \mathtt{a} \mid \mathtt{c} \mid A_0 \mid \cdots \mid A_n,$$
$$A_0 \to \mathtt{a}, \qquad A_i \to A_{i-1}A_{i-1} \mid \mathtt{c} \quad (1 \leq i \leq n)$$

as a fan-out-one MCFG. It is reduced and good, has size $O(n)$ and rule rank two, generates $\{\mathtt{a}, \mathtt{c}\}^+$, and every rule has a successful use of length at most two, so $\theta_{G_n} \leq 2$.

In the full refinement, the retained copy $(A_n)_1$ can derive only words without $\mathtt{c}$; recursively the unique such yield is $\mathtt{a}^{2^n}$. Hence the typed rule $S_{(1)} \to (A_n)_1$ has successful thickness at least 2^n. □

Thus full yield-type refinement admits no bound polynomial in $\|G\| \max\{1, \theta_G\}$ for arbitrary fixed typing, already at fan-out one and rank two. This is a limitation of the full-refinement route, not of the language $\{\mathtt{a}, \mathtt{c}\}^+$ itself.

Prior lower bound at fan-out one. The revised companion CFG paper [25] (whose frozen TCS R1 source is linked in the bibliography) establishes an operator-independent exponential characteristic-data lower bound for fixed two-element semilattice typing, already at fan-out one and rule rank two [25]. We do not reproduce this prior CFG result. It does not automatically extend to higher fan-out, whose tuple substitutability tests more arities.

4.3 Rule Rank One: Grammar-Size Polynomial Data

Unary derivation chains admit a grammar-size bound independent of thickness, even under arbitrary fixed componentwise typing.

Theorem 4.5 (Rank-one polynomial construction and characteristic data)**.** *Fix $f \geq 1$, finite Σ, and a finite-semigroup typing $s : \Sigma^+ \to \mathsf{S}$. Put $h = \widehat{s}$ and $M = \mathsf{S}^\sharp$. For any finite standard fan-out-$\leq f$, rank-≤ 1 grammar G_{in} of $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$, $\mathcal{B}^{\leq 1}_{f,h}$ is polynomial-time in $\|K\|_+$ and has a characteristic sample of total length $\mathrm{poly}_{f,s}(\|G_{\mathrm{in}}\|)$. The direct-start datum ε is added when $\varepsilon \in L$ without affecting this bound. The conservative learner $\mathcal{A}^{\leq 1}_{f,h}$ consequently TxtEx-identifies $\mathcal{C}^{\mathrm{mcf}}_{f,s} \cap \mathsf{MCFL}(f, 1)$ with polynomial time per datum.*

Proof. Normalize without increasing rank (Proposition 2.6), and trim the typed refinement. Write $n_{\mathrm{typ}} \leq |V||M|^f$ for its number of non-start typed nonterminals and $b_G \leq \max\{1, \|G\|\}$ for the largest explicit terminal count of a rule, taking $b_G \geq 1$. Choose a typed terminal unary derivation that minimizes first total yield length and then the number of rules. No typed nonterminal can repeat: splicing in the later suffix would not increase yield length and would strictly shorten the chain. Therefore $\|\omega(X)\|_1 \leq n_{\mathrm{typ}} b_G$. Lemma 4.2 bounds each witness word by $(2n_{\mathrm{typ}} + 1)b_G$, while Proposition 3.11 gives at most $n_{\mathrm{typ}} + |P||M|^f$ witnesses. Hence

$$\|\mathsf{W}(\widetilde{G}_0)\|_+ \leq (n_{\mathrm{typ}} + |P||M|^f)(1 + (2n_{\mathrm{typ}} + 1)b_G).$$

The polynomial overhead of normalization and the time and convergence bounds follow from Theorem 3.14 and Corollary 3.16. □

This extends the rank-one untyped scale [14, Lemma 7] and the fan-out-one argument of [25] to arbitrary fixed finite typing at every fixed fan-out. Rank two cannot generally yield a grammar-size-only bound: the size-$O(n)$ grammar $A_0 \to \mathtt{c}$, $A_i \to A_{i-1}A_{i-1}$, $S \to A_n$ generates the singleton $\{\mathtt{c}^{2^n}\}$, which is (f, h)-tuple-substitutable for every fixed f, h: over a one-letter alphabet, sharing a sentence context forces equal total component lengths, so every context accepts both tuples or neither. Every characteristic sample for this singleton target contains a word of length at least 2^n.

4.4 Prefix–Suffix-Determined Typings: Thickness-Sensitive Data

Lemma 4.3 gives an arbitrary-h characteristic-data bound in terms of the thickness of the refined typed presentation. Proposition 4.4 shows that this refined thickness need not be polynomially controlled by the original good presentation. A stronger transfer is nevertheless possible whenever the finite typing is determined by a fixed prefix–suffix window. We formulate this boundary-determined regime in the same thickness-sensitive representation model used by Yoshinaka [14, Section 4.3].

Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG and put $\bar{\theta}_G := \max\{1, \theta_G\}$. Here θ_G is the rule-thickness parameter defined above. The argument is stated for the λ-free good core. If the target also contains ε, apply the argument to $L \setminus \{\varepsilon\}$ and add the single datum ε at the start symbol, exactly as in Theorem 3.13; this adds only one length-zero example and does not affect either polynomial bound.

Lemma 4.6 (Rule thickness bounds local terminal material)**.** *Let G be reduced and good. Then:*

(i) for every rule R, the total number of terminal symbols written explicitly in the template of R is at most θ_G;

(ii) for every nonterminal A, there exists a terminal tuple $\vec{u} \in L_A(G)$ with $\|\vec{u}\|_1 \le \theta_G$.

Proof. Reducedness places every rule in some successful derivation. Because a good grammar is nondeleting, every terminal written by that rule and every component of the tuple produced at its left-hand side survives exactly once to the final sentence. A minimum-yield successful derivation using the rule therefore bounds both its explicit terminal material and the total length of that occurrence's left-hand-side tuple. Maximizing over rules gives (i) and (ii). □

Lemma 4.7 (Untyped rule thickness controls canonical contexts)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG and $\widetilde{G}_0$ its trimmed yield-type refinement for any fixed finite typing h. For each $A \in V$ of fan-out d there is an admissible sentence context $E_A \in \mathcal{E}_d$ such that*

$$|E_A|_{\mathrm{fix}} \le \bar{\theta}_G, \qquad E_A[\vec{u}] \in L(G) \quad \text{for all } \vec{u} \in L_A(G),$$

and $E_S = \square_1$ may be chosen. Thus for every non-start typed nonterminal $X = A_{\mathbf{t}}$ the choice $\chi(X) := E_A$ meets Definition 3.10. With canonical tuples satisfying $\|\omega(X)\|_1 \le b_{\mathrm{can}}$ and maximum rule rank $r_{\max}$, every word in $\mathsf{W}(\widetilde{G}_0)$ has length at most $2\bar{\theta}_G + r_{\max} b_{\mathrm{can}}$.

Proof. Take $E_S = \square_1$. For $A \neq S$ choose a source rule R with left-hand side A. Because G is reduced, there is a successful sentence derivation using R and having length $\theta_G(R) \le \theta_G$. Remove the subtree rooted at that occurrence of A, replacing its d components by ordered holes. Lemma 2.11, iterated along the path to the root, and goodness show that the resulting E_A belongs to $\mathcal{E}_d$. Its fixed terminal material is part of the selected sentence, hence has length at most $\theta_G(R)$. Substituting any tuple from $L_A(G)$ gives another successful source derivation, so E_A accepts every such tuple.

By Proposition 3.6(i), $L_X \subseteq L_A(G)$; consequently E_A accepts every tuple of the typed X. The surrounding source derivation can also be lifted to the full typed refinement by Proposition 3.6(ii): the type of X determines all ancestral types given the fixed sibling derivations, and the fresh start symbol admits every start type. In particular, $\chi(X)$ has the required common-context property. For a direct start link, $X = S_{\mathbf{t}}$ and the chosen context is $\square_1$.

A witness is $\chi(X)[\omega(X)]$, $\chi(X)[\vec{c}]$, or $\chi(X)[\rho(\omega(Y_1), \ldots, \omega(Y_r))]$. Its context contributes at most $\bar{\theta}_G$ terminals; the source rule writes at most $\bar{\theta}_G$ by Lemma 4.6(i), and it has at most $r_{\max}$ child tuples. The displayed bound follows, with the rank-zero case included. □

Fix $k, \ell \geq 0$ and let $h : \Sigma^* \to M$ be any finite typing with $h \preceq h_\Sigma^{k,\ell}$. Thus h factors through the surjective prefix–suffix typing $h_\Sigma^{k,\ell}$. Let G have fan-out at most f and rule rank at most a fixed $r_{\max} \geq 1$, and let $\widetilde{G}_0$ be its trimmed full yield-type refinement under h. Let n_{typ} be the number of retained non-start h-typed nonterminals.

Lemma 4.8 (Short prefix–suffix-typed canonical tuples)**.** *For every non-start typed nonterminal X of $\widetilde{G}_0$, the canonical tuple may be chosen so that $\|\omega(X)\|_1 \leq f(k+\ell+1)n_{\mathrm{typ}}(r_{\max}+1)\bar{\theta}_G$.*

Proof. Choose, among all terminal derivations from X in the trimmed typed grammar $\widetilde{G}_0$, one with the minimum number of grammar-rule nodes, and then mark the root terminals needed to determine its componentwise $h_\Sigma^{k,\ell}$-type: all terminals of a component shorter than $m_0 = \max\{1, k+\ell\}$ from Definition 2.4, and otherwise its first k and last ℓ terminals (one arbitrary terminal when $k = \ell = 0$). There are at most $f(k+\ell+1)$ marked terminals.

No typed nonterminal can occur twice on a root-to-leaf grammar path of this minimal derivation. If an upper and a proper lower occurrence had the same typed label, the lower subtree would itself be a valid typed terminal derivation from the same typed nonterminal, and replacing the upper subtree by it would strictly reduce the number of grammar-rule nodes. Every typed nonterminal occurring in this derivation is retained in $\widetilde{G}_0$; hence every root-to-marked-terminal path contains at most n_{typ} grammar nodes. Their union therefore contains at most $f(k+\ell+1)n_{\mathrm{typ}}$ grammar nodes.

Keep the union of these marked root-to-leaf paths as the retained subderivation. By linearity and nondeletion, each component of every omitted child contributes a contiguous interval to exactly one root component; it is never split between root components. Replace every omitted child subtree by a shortest terminal tuple of its underlying untyped nonterminal. Lemma 4.6(ii) bounds each such tuple by $\bar{\theta}_G$, and part (i) bounds the explicit terminal material of each retained rule by $\bar{\theta}_G$. Because the grammar is good, every replacement component is nonempty. An omitted child cannot contribute to a short root component: every terminal of such a component was marked, so any nonempty contribution from that child would contain a marked terminal and the corresponding root-to-terminal path would place the child on the retained skeleton. In a long root component no replaced block crosses a protected prefix or suffix position. Hence every replacement lies wholly in the unmarked interior of that component: the marked first k and last ℓ terminals remain literal prefixes and suffixes, and the component stays on the long-word side of the threshold. Thus the root prefix–suffix type is unchanged (and when $k = \ell = 0$ the single marked terminal preserves nonemptiness). Applying the factor map supplied by $h \preceq h_\Sigma^{k,\ell}$ shows that its componentwise h-type is unchanged as well. Each retained node contributes at most $\bar{\theta}_G$ explicit terminals and has at most $r_{\max}$ omitted child subtrees. The resulting root tuple has length at most $f(k+\ell+1)n_{\mathrm{typ}}(r_{\max}+1)\bar{\theta}_G$. It therefore has the h-type indexing X. The replacements may change internal typed labels of the original typed derivation; this is harmless. We retain only the underlying G-derivation and the preserved root type, and then Proposition 3.6(ii) uniquely lifts that derivation back to G^h with root X. Since X belongs to $\widetilde{G}_0$, the lifted derivation can be inserted into a successful derivation through X and hence survives trimming. Its root tuple may therefore be chosen as $\omega(X)$. □

Apply Lemma 4.7 with $b_{\mathrm{can}} = f(k+\ell+1)n_{\mathrm{typ}}(r_{\max}+1)\bar{\theta}_G$. Every selected witness word then has length at most $2\bar{\theta}_G + r_{\max}b_{\mathrm{can}} = O_{f,r_{\max},k,\ell,h,\Sigma}(\|G\|\bar{\theta}_G)$, since $n_{\mathrm{typ}} \leq |V||M|^f$.

Theorem 4.9 (Polynomial data for prefix–suffix-determined typings)**.** *Fix $f \geq 1$, $r_{\max} \geq 1$, k, ℓ, a finite alphabet Σ, and a finite-semigroup typing $s : \Sigma^+ \to \mathsf{S}$ satisfying $s \preceq s_\Sigma^{k,\ell}$; put $h = \widehat{s}$. Let $L \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$ and let G be a reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ grammar with $L(G) = L \cap \Sigma^+$. Then $\mathcal{B}_{f,h}^{\leq r_{\max}}$ is polynomial-time in the positive sample. A characteristic sample*

for L is $K = \mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$, with

$$|\mathsf{W}(\widetilde{G}_0)| = O_{f,r_{\max},s}(\|G\|), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},k,\ell,s,\Sigma}(\|G\|^2 \bar{\theta}_G),$$

where $\bar{\theta}_G = \max\{1, \theta_G\}$. For a reduced λ-free input presentation G_{in} (deleting rules allowed), the structural good-form conversion of Proposition A.1 yields the explicit presentation-relative bound

$$O_{f,r_{\max},k,\ell,s,\Sigma}\Big(\|G_{\mathrm{in}}\|^3 \max\{1, t_{G_{\mathrm{in}}}\}\Big).$$

Proof. The h-refinement has at most $|V||M|^f$ typed nonterminals and $|P||M|^{fr_{\max}}$ typed rule instances. Lemma 4.8 applies because preserving the finer $h^{k,\ell}_\Sigma$ boundary type preserves the coarser h-type. Lemma 4.7 now bounds every selected witness word by $O_{f,r_{\max},k,\ell,h,\Sigma}(\|G\|\bar{\theta}_G)$. Proposition 3.11 gives $|\mathsf{W}(\widetilde{G}_0)| \le n_{\mathrm{typ}} + |P||M|^{fr_{\max}}$, with $n_{\mathrm{typ}} \le |V||M|^f$. Multiplying the polynomial number of witnesses by the preceding word-length bound, and adding at most one datum for ε, proves the asserted total-positive-length bound. The same successful witnesses also imply a polynomial bound on the refined typed rule thickness, but that intermediate parameter is not needed for the estimate. Theorem 3.14 gives the hypothesis-construction claim. □

The input-presentation clause follows from Proposition A.1; nonterminal thickness is used because deleting source rules may erase material before good-form conversion. Corollary 3.16 then gives polynomial time per processed datum for the corresponding capped conservative learner.

At fan-out one, this polynomial boundary-typing bound is not new: Yoshinaka's learner for (k,ℓ)-substitutable context-free languages already yields polynomial characteristic data in grammar size and thickness [12, Theorem 1], and the revised companion CFG manuscript extends the window result to finite-group refinements [25]. The contribution of Lemma 4.8 is to preserve boundary information simultaneously across *multiple* tuple components, giving the same kind of bound at every fixed fan-out $f \ge 2$.

The bound is relative to a reduced good bounded-rank presentation. Theorem 5.3 below shows that this higher-rank regime is needed at the language level: it gives a non-context-free $(1,1)$-typed target of minimum fan-out two and minimum rule rank two even when competing presentations may use arbitrary finite fan-out. Of the examples below, L_{intr} falls under the window theorem, whereas the rank-one languages L_{cd} and $L_\star$ fall under Theorem 4.5; thus each separation is compatible with a quantitative learning bound.

4.5 An Algebraic Criterion for Polynomial Characteristic Data

Write **LI** for the pseudovariety of finite locally trivial semigroups and **Grp** for the pseudovariety of finite groups. Their join **LI** ∨ **Grp** consists of divisors of finite products $S \times A$, with $S \in$ **LI** and A a finite group [28, 29]. The following criterion makes the window–group hypothesis intrinsic to the positive image of the fixed typing. Its finite-semigroup ingredient also occurs in the companion CFG manuscript [25].

Lemma 4.10 (Locally trivial–group typings admit window–group refinements)**.** *Let Σ be a nonempty finite alphabet and let $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ be a finite-semigroup typing; put $h = \widehat{s}$. If $\mathsf{S} \in$ **LI** ∨ **Grp**, then there exist $k, \ell \ge 0$, a finite group A, and a morphism $g : \Sigma^* \to A$ such that*

$$h \preceq (h^{k,\ell}_\Sigma, g) \quad \text{on all of } \Sigma^*.$$

*If $\mathsf{S} \in$ **LI**, the group factor can be taken trivial, so $h \preceq h^{k,\ell}_\Sigma$.*

Proof. Choose a finite locally trivial semigroup S, a finite group A, a subsemigroup $T \subseteq S \times A$, and an onto semigroup morphism $\pi : T \to \mathsf{S}$. For each $a \in \Sigma$ lift $s(a)$ to a pair $(s_a, g_a) \in T$. Multiplying these pairs extends the lifts to a morphism $\rho : \Sigma^+ \to T$ with $\pi\rho = s$. Let $\sigma : \Sigma^+ \to S$ and $g : \Sigma^* \to A$ denote the two coordinate morphisms, with $g(\varepsilon) = 1_A$.

The positive image $S_0 = \sigma(\Sigma^+)$ is locally trivial. Put $q = |S_0| + 1$. The standard finite locally trivial identities $S_0^q = S_0 E(S_0) S_0$ and $euf = ef$ for idempotents $e, f \in S_0$ and $u \in S_0$ [25, 27] imply that $\sigma(w)$ is determined by its length-q prefix and suffix whenever $|w| \geq 2q$: writing $w = pxz$ with $|p| = |z| = q$, choose $\sigma(p) = aeb$ and $\sigma(z) = cfd$; writing $\sigma^1(\varepsilon) = 1$ in the adjoined identity monoid, we obtain

$$\sigma(pxz) = ae\,(b\,\sigma^1(x)\,c)\,fd = aefd = \sigma(pz).$$

The parenthesized factor belongs to S_0 even if $x = \varepsilon$, because $b, c \in S_0$. Words shorter than $2q$ are recorded individually by $h_\Sigma^{q,q}$. Therefore equality of $h_\Sigma^{q,q}$-values on Σ^+ implies equality of σ-values. Equality of both window and g values then implies equality of ρ values and hence of h values.

The window typing distinguishes ε from every nonempty word, so the kernel inclusion extends to all of Σ^*. Taking $k = \ell = q$ proves the first assertion. If S itself is locally trivial, use $S = \mathsf{S}$, its identity projection for π, and the trivial group. □

Corollary 4.11 (Algebraic sufficient conditions for polynomial data)**.** *Fix a nonempty finite Σ, a fan-out cap $f \geq 1$, a rule-rank cap $r_{\max} \geq 1$, and a finite-semigroup typing $s : \Sigma^+ \to \mathsf{S}$; put $h = \widehat{s}$. Let G be any reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ presentation of a target in $\mathcal{C}^{\mathrm{mcf}}_{f,s}$, and put $\bar{\theta}_G = \max\{1, \theta_G\}$. The capped reconstruction operator has a characteristic sample of cardinality $O_{f,r_{\max},h}(\|G\|)$ with:*

$$\|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},h,\Sigma}(\|G\|^2 \bar{\theta}_G), \quad \text{if } \mathsf{S} \in \mathbf{LI},$$
$$\|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},h,\Sigma}(\|G\|^3 \bar{\theta}_G), \quad \text{if } \mathsf{S} \in \mathbf{LI} \vee \mathbf{Grp}.$$

Proof. For **LI**, apply Lemma 4.10 with trivial group and Theorem 4.9. For $\mathbf{LI} \vee \mathbf{Grp}$, use the finite window–group refinement from that lemma and Theorem C.7. The parameters of the refining morphisms depend only on the fixed typing and alphabet, not on G. The bound on cardinality holds in both cases. □

Remark 4.12. These are *sufficient* algebraic conditions, not a characterization of all typings admitting polynomial original-thickness data. The lower-bound question beyond fan-out one in Open Problem 6.1 remains open. The empty-alphabet case has only the word ε and is trivial.

4.6 Computing Short Characteristic Data

The bounds above are existential: each proof exhibits a short tuple or context. The shortest ones can be computed directly. Call a sentence context of a typed nonterminal X *derivational* if it arises, as in the proof of Lemma 3.8, from a successful typed derivation by deleting one occurrence of the subtree rooted at X. Every such context is admissible for X. The contexts E_A of Lemma 4.7, lifted to $X = A_{\mathbf{t}}$, and those of Lemma 4.2 are derivational.

Proposition 4.13 (Effective short characteristic data)**.** *Fix $f, r_{\max}$ and a finite typing $h : \Sigma^* \to M$. Given a reduced good MCFG G of fan-out at most f and rule rank at most $r_{\max}$, the following can be computed in time polynomial in $\|G\|$, with lengths written in binary:*

(i) $\theta_G(R)$ for every rule R, and hence θ_G and t_G;

(ii) the trimmed refinement $\widetilde{G}_0$ and, for each non-start typed nonterminal X, a canonical tuple $\omega(X) \in L_X$ of minimum total length and a derivational context $\chi(X)$ of minimum fixed length, with $\chi(X) = \square_1$ whenever $\widetilde{S} \to X$ belongs to $\widetilde{G}_0$;

(iii) derivation DAGs of size polynomial in $\|G\|$ representing these tuples and contexts, and hence every word of the witness set $\mathsf{W}(\widetilde{G}_0)$ of Definition 3.10 built from them.

Every word-length and total-length bound proved in this section and in Appendices B, C and D holds for this computed witness set. Its words can be written out explicitly in time polynomial in $\|G\| + \|\mathsf{W}(\widetilde{G}_0)\|_+$.

Proof. For fixed $f, r_{\max}, h$ the full refinement G^h has at most $|V||M|^f$ typed nonterminals and $|P||M|^{fr_{\max}}$ typed rules, and trimming is a linear-time productivity and reachability computation. View the typed rules as weighted deduction steps. A rule $X \to \rho(Y_1, \dots, Y_r)$ maps child lengths x_j to $c_\rho + \sum_j x_j$, where c_ρ is the number of terminals written by ρ; this function is superior. To break zero-length ties without cyclic predecessor pointers, minimize the lexicographic pair (terminal length, derivation-node count), adding $(c_\rho, 1)$ at each rule. The pair-valued operation is strictly superior to each child even when no terminal is written. The shortest-derivation algorithm in the weighted deductive form of Nederhof [36], building on Knuth's generalization of Dijkstra's method for the grammar problem (see also Ramalingam and Reps [35]), therefore computes the minimum length $\lambda(X) = \min\{\|\vec{u}\|_1 : \vec{u} \in L_X\}$ of every typed nonterminal together with an acyclic choice of minimizing rules. These choices form a derivation DAG for $\omega(X)$, with shared subderivations. A second Dijkstra pass computes outside lengths, breaking equal-length ties by the number of edges. It starts with $\mu(X) = 0$ when $\widetilde{S} \to X$ is present. For every typed rule $X \to \rho(Y_1, \dots, Y_r)$ and every j it adds an edge from X to Y_j of weight $c_\rho + \sum_{i \neq j} \lambda(Y_i)$. The value $\mu(X)$ is the minimum fixed length of a derivational context of X, and the shortest-path tree supplies its DAG. The same two passes on G itself give $t_G = \max_A \lambda(A)$ and $\theta_G(R) = \mu(A) + c_\rho + \sum_j \lambda(B_j)$ for $R : A \to \rho(B_1, \dots, B_r)$.

Each proof in this section and in the cited appendices bounds a witness word by bounding the length of its context, its rule template, and its canonical child tuples separately. It does so by exhibiting some tuple in L_X and some derivational context of length within the stated bound. The computed minima are no longer, so every stated bound persists. Expanding a DAG takes time linear in the length of the word it represents. □

Proposition 4.14 (Rule-only characteristic witnesses)**.** *In the setting of Proposition 4.13, choose all canonical tuples by the consistent acyclic minimum-weight derivations described there. For each non-start typed nonterminal X, let $R_X : X \to \rho_X(Y_1, \dots, Y_q)$ be the first rule in its chosen minimizing derivation (including $q = 0$). Then*

$$\omega(X) = \rho_X(\omega(Y_1), \dots, \omega(Y_q)).$$

Consequently the witness set of Definition 3.10 equals the set of rule-witness words alone:

$$\mathsf{W}(\widetilde{G}_0) = \{\chi(\operatorname{lhs} R)[\rho_R(\omega(Y_1), \dots, \omega(Y_q))] : R \in \mathcal{R}\},$$

where $\mathcal{R}$ denotes the non-start typed rules and rank-zero rules are included. In particular,

$$|\mathsf{W}(\widetilde{G}_0)| \leq |\mathcal{R}| \leq |P||M|^{fr_{\max}},$$

for a fixed typing and rule-rank cap. The set remains characteristic for $\mathcal{B}_{f,h}^{\leq r_{\max}}$ whenever the target is in $\mathcal{C}_{f,h}^{\mathrm{mcf}}$.

Proof. For each typed nonterminal X, an optimal derivation starts with a rule R_X. Every child derivation must have minimum weight: replacing a nonoptimal child by the chosen minimum would decrease the lexicographic weight of the whole derivation. The additional derivation-node tie-breaker makes the simultaneous choice of minimizing rules acyclic. Hence the displayed identity for $\omega(X)$ holds, also for rank zero.

It follows that $\chi(X)[\omega(X)]$ is *identical* to the witness word of R_X. Every other term in Definition 3.10 is already a rule witness. Thus the two sets coincide as sets of terminal words; in particular, the completeness and characteristic-sample arguments using $\mathsf{W}(\widetilde{G}_0)$ are unchanged. There is at most one selected word per rule instance. The refinement bound on the number of typed rules gives the displayed cardinality estimate. □

Proposition 4.15 (A computable presentation-sensitive data budget)**.** *Under the hypotheses of Proposition 4.13, let $\mathcal{R}$ be the non-start typed rules of $\widetilde{G}_0$. Let $\lambda(X)$ be the minimum total yield length from X, and let $\mu(X)$ be the minimum fixed length of a derivational sentence context of X (with $\mu(X) = 0$ for a direct start link). For $R : X \to \rho(Y_1, \dots, Y_q)$, put*

$$b(R) := c_\rho + \sum_{j=1}^{q} \lambda(Y_j), \qquad \mathsf{B}_{\mathrm{rule}}(G, h) := \sum_{R \in \mathcal{R}} \big(1 + \mu(\mathrm{lhs}\, R) + b(R)\big).$$

There is a computable witness set $\mathsf{W}(\widetilde{G}_0)$ with

$$\|\mathsf{W}(\widetilde{G}_0)\|_+ \le \mathsf{B}_{\mathrm{rule}}(G, h).$$

The integer $\mathsf{B}_{\mathrm{rule}}(G, h)$ is computable in polynomial time in $\|G\|$ when $f, r_{\max}, h$ are fixed, with its value represented in binary. If $L(G)$ also contains the empty word and its nonempty core is treated separately, the optional ε datum adds at most one to the encoded budget.

Proof. Use the minimizing canonical tuples and derivational contexts constructed in Proposition 4.13. By Proposition 4.14, every canonical-nonterminal witness is already one of the rule witnesses. For each positive-rank or rank-zero typed rule $R : X \to \rho(Y_1, \dots, Y_q)$, its witness word $\chi(X)[\rho(\omega(Y_1), \dots, \omega(Y_q))]$ has length exactly $\mu(X) + b(R)$. Summing the encoded lengths $|w| + 1$ over this indexed multiset of rule witnesses gives $\mathsf{B}_{\mathrm{rule}}(G, h)$. Passing to the set of distinct words can only lower the total length. The computations of λ, μ, the number of typed symbols and rules, and the binary lengths of these quantities are polynomially bounded as in Proposition 4.13. □

Remark 4.16. This *rule-only* budget is presentation-sensitive and can be much smaller than substituting worst-case maxima into the polynomial bounds. It is an efficiently computable *upper certificate*, not a claim of a minimum-size characteristic sample or an improved worst-case polynomial exponent.

The singleton $\{\mathsf{c}^{2^n}\}$ after Theorem 4.5 shows that explicit characteristic words can be exponentially longer than $\|G\|$. In that case the derivation DAGs of Proposition 4.13 remain of polynomial size, so the computed sample has a polynomial-size compressed description.

4.7 Distributional Representatives and Abelian-Filtered Targets

The computable rule-only budget above uses representatives *generated by* the typed grammar. For an existential characteristic-sample bound there is a different option: choose short representatives of the same *typed distribution classes*, even when the representatives themselves have no derivation from the corresponding nonterminal.

Proposition 4.17 (Relaxed distributional witnesses)**.** *Let G be a reduced good fan-out-$\le f$, rank-$\le r$ presentation of $L = L(G) \in \mathcal{C}^{\mathrm{mcf}}_{f,h}$, and let $\widetilde{G}_0$ be its trimmed yield-type refinement. For every non-start typed nonterminal X, let $[X]$ be the $\equiv_{L,h}$-class containing L_X, and put*

$$\kappa_G(X) = \min_{\vec{v} \in [X]} \|\vec{v}\|_1, \qquad \kappa_G = \max_X \kappa_G(X).$$

There exists a characteristic positive sample $K^\dagger$ for both the capped operator $\mathcal{B}^{\le r}_{f,h}$ (with cap at least the rank of G) and the uncapped $\mathcal{B}_{f,h}$ satisfying

$$|K^\dagger| \le |V||M|^f + |P||M|^{fr},$$
$$\|K^\dagger\|_+ \le (|V||M|^f + |P||M|^{fr})(1 + \theta_G + \|G\| + (r+1)\kappa_G).$$

The bound concerns the nonempty core; if $\varepsilon \in L$, add the sample word ε at encoded cost one.

Proof. For each $X = A_{\mathbf{t}}$ choose a tuple $\omega^\dagger(X) \in [X]$ of total length $\kappa_G(X)$. Choose a derivational sentence context $\chi^\dagger(X)$ for the *untyped* A using a shortest successful derivation containing an occurrence of A, with all sibling yields fixed. The fixed material of this context is at most θ_G, since any such successful derivation witnesses some rule involving A; if $\widetilde{S} \to X$ is a start link, use $\square_1$ instead. This context accepts a tuple $u \in L_X$. As $\omega^\dagger(X) \equiv_{L,h} u$, it also accepts $\omega^\dagger(X)$.

Form $K^\dagger$ from all words $\chi^\dagger(X)[\omega^\dagger(X)]$, all rule words $\chi^\dagger(X)[\rho(\omega^\dagger(Y_1), \dots, \omega^\dagger(Y_q))]$, and all rank-zero rule witnesses. By Lemma 3.3, the rule tuple in the second family is distributionally equivalent to a tuple in L_X, so every listed word belongs to L. The proof of Proposition 3.12 now works with $\iota(X) = [\omega^\dagger(X)]$: the chosen contexts supply the Link instances, while the rule words expose the required composition witnesses. Proposition 3.4 gives the reverse inclusion for arbitrary larger positive samples.

There is one word per typed nonterminal and one per typed rule before deduplication. A nonterminal witness has length at most $\theta_G + \kappa_G$, and a rule witness at most $\theta_G + \|G\| + r\kappa_G$. Summing the encoded costs gives the bound. Unlike Proposition 4.14, these representatives need not be generated by their nonterminals, so the nonterminal words cannot in general be omitted. □

Remark 4.18. For fixed f, r, h, the preceding bound implies that any family G_n with $\|G_n\| \max\{1, \theta_{G_n}\} = n^{O(1)}$ and superpolynomial characteristic-data requirements must satisfy $\kappa_{G_n} = n^{\omega(1)}$. This is a necessary condition, not a proof that such a family exists. The relaxed construction is existential; the rule-only algorithm of Proposition 4.15 is a distinct, effective construction using grammar-generated representatives.

Theorem 4.19 (Polynomial data for abelian-filtered targets)**.** *Fix a finite nonempty alphabet Σ, a finite semigroup typing $s : \Sigma^+ \to \mathsf{S}$ and bounds $f, r \ge 1$. Put $h = \widehat{s}$ and $M = \mathsf{S}^\sharp$. There are constants C, e depending only on Σ, s, f, r with the following property. Let $\mathbb{A}$ be any abelian group, $\varphi : \Sigma^* \to \mathbb{A}$ a morphism, $\mathcal{H} \le \mathbb{A}$, $\gamma \in \mathbb{A}$, and $\mathsf{Acc} \subseteq \mathsf{S}^\sharp$. If*

$$L = h^{-1}(\mathsf{Acc}) \cap \varphi^{-1}(\gamma + \mathcal{H})$$

has a reduced good fan-out-$\le f$, rank-$\le r$ MCFG presentation G of $L \cap \Sigma^+$, then $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and admits a characteristic sample for $\mathcal{B}^{\le r}_{f,h}$ with

$$\|K^\dagger\|_+ \le C(\|G\| \max\{1, \theta_G\})^e.$$

The constants do not depend on $\mathbb{A}, \varphi, \mathcal{H}, \gamma$. The datum ε is included when $\varepsilon \in L$; the degenerate cases $L \subseteq \{\varepsilon\}$ are handled directly.

Proof. Membership in $\mathcal{C}^{\mathrm{mcf}}_{f,s}$ follows from Proposition 2.10; the positive part has the assumed MCFG presentation, and ε is admitted separately by the Start rule.

Let $\psi : \Sigma^* \to \mathbb{N}^\Sigma$ be the Parikh map and write $\Psi(\vec{u}) = \sum_i \psi(u_i)$. The morphism φ factors through a group homomorphism $\overline{\varphi} : \mathbb{Z}^\Sigma \to \mathbb{A}$. Put

$$\Lambda = \{z \in \mathbb{Z}^\Sigma : \overline{\varphi}(z) \in \mathcal{H}\}.$$

Fix for each untyped nonterminal A a productive tuple ω_A of length at most the untyped nonterminal thickness $t_G \le \theta_G$. Since any two A-tuples can be inserted into the same successful derivational context and their resulting words belong to L,

$$\Psi(u) - \Psi(\omega_A) \in \Lambda \qquad (u \in L_A(G)). \tag{$*$}$$

For each rule $R : A \to \rho(B_1, \dots, B_q)$, let $c_R \in \mathbb{N}^\Sigma$ count its explicit terminal letters and define

$$\delta_R = \Psi(\omega_A) - c_R - \sum_{j=1}^{q} \Psi(\omega_{B_j}).$$

By (*) this vector lies in Λ, and its ℓ_1-norm is bounded by $T = (r+1)t_G + \|G\|$. Let $\Gamma_T = \{z \in \Lambda : \|z\|_1 \le T\}$ and $\Lambda_T = \langle \Gamma_T \rangle$. Induction on a derivation, using

$$\Psi(u) - \Psi(\omega_A) = \sum_j (\Psi(u_j) - \Psi(\omega_{B_j})) - \delta_R,$$

shows that every $u \in L_A(G)$ satisfies $\Psi(u) - \Psi(\omega_A) \in \Lambda_T$.

Consider now any typed nonterminal $X = A_{\mathbf{t}}$ with $d \le f$, and fix $u = (u_1, \dots, u_d) \in L_X$. For each component choose one linear set from a fixed semilinear representation of the Parikh image of the regular language $h^{-1}(t_i) \cap \Sigma^+$ containing $\psi(u_i)$. Write that linear set as $c_i + \sum_{p \in P_i} \mathbb{N}p$. The number and coefficients of all these generators are bounded by constants depending only on Σ and h. The nonnegative integer system

$$\sum_{i=1}^{d} \left(c_i + \sum_{p \in P_i} \nu_{i,p} p \right) - \sum_{z \in \Gamma_T} (\mu^+_z - \mu^-_z) z = \Psi(\omega_A)$$

is feasible: use the Parikh vectors of the original u_i and the preceding Λ_T-membership. It has exactly $|\Sigma|$ equations, at most $O_{f,h,\Sigma}((T+1)^{|\Sigma|})$ nonnegative variables, and coefficients of absolute value $O_{f,h,\Sigma}(T+1)$. Papadimitriou's small nonnegative integral solution bound [41] therefore gives a solution with every coefficient bounded by a fixed polynomial in $T+1$. For each resulting Parikh vector select a nonempty word $v_i \in h^{-1}(t_i)$ realizing it, and let $\vec{v} = (v_1, \dots, v_d)$. Then $\|\vec{v}\|_1$ is polynomial in $T+1$, the componentwise h-types agree with u, and $\Psi(\vec{v}) - \Psi(u) \in \Lambda_T \subseteq \Lambda$.

Finally, for any sentence context E, the total h-type of $E[\vec{v}]$ agrees with that of $E[u]$, and

$$\varphi(E[\vec{v}]) - \varphi(E[u]) = \overline{\varphi}(\Psi(\vec{v}) - \Psi(u)) \in \mathcal{H}.$$

Hence $\vec{v} \equiv_{L,h} \vec{u}$. This constructs a representative of every typed distribution class with length polynomial in $T+1$. Applying Proposition 4.17 and $t_G \le \theta_G$ yields the claimed total-length bound, with parameters independent of the auxiliary abelian group. □

Remark 4.20. Theorem 4.19 applies even when the positive image of h is *outside* **LG**. It is a sufficient condition on the target language, rather than on the finite control monoid. It does not claim polynomial-time construction for arbitrary, possibly ineffectively presented auxiliary abelian groups.

5 Comparison with Multidimensional Substitutability

Yoshinaka's multidimensional substitutability is the direct positive-data baseline [13, 14]. Under trivial typing the present framework reproduces that semantic class; finite typing refines which observed tuples may be identified while preserving compositional rule witnesses. The MCFG proofs additionally require componentwise type propagation and, for the quantitative boundary theorem, simultaneous control of several tuple components. Yoshinaka's (k, ℓ) hierarchy is a separate CFG result [12]; $h_\Sigma^{k,\ell}$ lifts the same boundary information componentwise. Since every (f, h) condition includes arity one, $\mathcal{C}^{\mathrm{mcf}}_{f,h} \cap \mathrm{CFL} \subseteq \mathcal{C}^{\mathrm{mcf}}_{1,h}$. The boundary-typed example below exhibits intrinsic rule rank two.

5.1 Yoshinaka's Baseline and Prefix–Suffix Specialization

Definition 5.1 (Yoshinaka multidimensional substitutability)**.** Using the sentence-context sets $\mathcal{E}_d$ defined in Section 2, a language L is *fD-substitutable* if, for every $1 \le d \le f$, all $E, F \in \mathcal{E}_d$, and all $\vec{x}, \vec{y} \in (\Sigma^+)^d$, $E[\vec{x}], \quad E[\vec{y}], \quad F[\vec{x}] \in L \Longrightarrow F[\vec{y}] \in L$. Write $\mathsf{YSub}(f)$ for this semantic class and $\mathsf{YSL}(f, r) := \mathsf{YSub}(f) \cap \mathsf{MCFL}(f, r)$, where r bounds rule rank. We refer to the displayed four-instance implication as the *substitution implication*. Yoshinaka's original definition quantifies over every arity $d \le f$, uses nonempty tuple components, and requires nonempty internal separators in d-contexts; footnote 3 separately discusses the stronger alternative that permits empty components and separators [14, Section 3.1 and footnote 3].

Recall the finite monoid morphism $h_\Sigma^{k,\ell}$ from Definition 2.4. In Yoshinaka's word-level definition, $w_1, y_1, w_1', w_2, y_2, w_2' \in \Sigma^*$, $u \in \Sigma^k$, and $v \in \Sigma^\ell$, with $uy_1v, uy_2v \neq \varepsilon$; the condition is

$$w_1uy_1vw_1',\ w_1uy_2vw_1' \in L \quad \Longrightarrow \quad (w_2uy_1vw_2' \in L \iff w_2uy_2vw_2' \in L)$$

[12]. Thus the compared fragments uy_iv are nonempty even though the middle words y_i themselves may be empty. Equality of their $h_\Sigma^{k,\ell}$-values is exactly equality of the boundary information used in this condition, including the exact identity of words shorter than $m_0 = \max\{1, k + \ell\}$.

Specializations of the semantic condition. The trivial typing removes the componentwise type premise, so $\mathcal{C}^{\mathrm{mcf}}_{f,h_{\mathrm{triv}}} = \mathsf{YSub}(f) \cap f\text{-MCFL} \subseteq \mathcal{C}^{\mathrm{mcf}}_{f,h}$. At fan-out one, $(1, h_\Sigma^{k,\ell})$-substitutability is precisely Yoshinaka's word-level (k, ℓ) condition [12]: equal window types mean identical short words or common prefixes and suffixes of lengths k and ℓ, respectively. The same correspondence applies componentwise at larger fan-out.

Compatibility with Yoshinaka's efficiency scale. For trivial typing no productive nonterminal has more than one yield type. Yoshinaka's untyped grammar witnesses already satisfy $|K_G| \le |P|$ and $\|K_G\| \le |P|\tau_G$ [14, Lemma 7], with polynomial time and data for fixed dimension and rank [14, Theorem 1]. Thus the present quantitative claims concern new typed, multi-component settings rather than re-proving the trivial-typing result.

A boundary-typing fact used in the separation. A language is *strictly 2-local* if membership is determined by permitted adjacent pairs in $\vdash w \dashv$, for fixed endmarkers. The following elementary fact is needed in the intrinsic-rank example.

Lemma 5.2 (Strictly 2-local constraints are boundary typed)**.** *If $L_{\mathrm{sl}} \subseteq \Sigma^*$ is strictly 2-local, then for every $f \ge 1$, $L_{\mathrm{sl}} \in \mathcal{C}^{\mathrm{mcf}}_{f,h_\Sigma^{1,1}}$.*

Proof. Let equal-$h_{\Sigma}^{1,1}$-type nonempty tuples $\vec{x}, \vec{y}$ share an accepting context E_0. Internal adjacent pairs of each y_i are permitted because they occur in $E_0[\vec{y}]$. If $E[\vec{x}] \in L_{\mathrm{sl}}$, every adjacent pair of $E[\vec{y}]$ is either in fixed material, internal to a y_i, or crosses a component boundary. These are permitted respectively by $E[\vec{x}]$, by $E_0[\vec{y}]$, or by equality of the first/last letters encoded by the boundary type; end-marker pairs are unchanged. Thus $E[\vec{y}] \in L_{\mathrm{sl}}$, and symmetry finishes the proof. □

5.2 An Intrinsically Rank-Two Boundary-Typed Target

The next example makes the distinction between the rank-one and higher-rank quantitative regimes intrinsic to the language. Its rule-rank obstruction is not created by finite typing: the balanced-count factor used below already belongs to Yoshinaka's trivial-typing class. The construction instead combines that obstruction with minimum fan-out two and a separation from the trivial-typing class inside one boundary-typed target.

Let

$$\Sigma_{\mathrm{intr}} := \{\mathtt{a}, \mathtt{b}, \mathtt{l}, \mathtt{r}, \mathtt{p}, \mathtt{q}, \mathtt{c}, \mathtt{d}, \#, \$, \mathtt{u}, \mathtt{v}, \mathtt{w}\}$$

and put

$$P_{\mathrm{intr}} := \{\mathtt{lpc}, \mathtt{lpd}, \mathtt{lqc}, \mathtt{rpc}, \mathtt{rpd}\}.$$

Define

$$\mathrm{MIX}_2^+ := \{x \in \{\mathtt{a}, \mathtt{b}\}^+ : |x|_{\mathtt{a}} = |x|_{\mathtt{b}}\}, \qquad T_3^+ := \{\mathtt{u}^n\mathtt{v}^n\mathtt{w}^n : n \geq 1\},$$

and

$$L_{\mathrm{intr}} := P_{\mathrm{intr}} \# \mathrm{MIX}_2^+ \$ T_3^+.$$

To make the role of the witness explicit, let

$$\varphi_{\mathrm{mix}} : \{\mathtt{a}, \mathtt{b}\}^* \to \mathbb{Z}, \qquad \varphi_{\mathrm{mix}}(\mathtt{a}) = 1, \quad \varphi_{\mathrm{mix}}(\mathtt{b}) = -1.$$

Then $\mathrm{MIX}_2^+ = \{\mathtt{a}, \mathtt{b}\}^+ \cap \varphi_{\mathrm{mix}}^{-1}(0)$. The language $\{\mathtt{a}, \mathtt{b}\}^+$ is trivially f-tuple-substitutable, so Proposition 2.10(ii) gives $\mathrm{MIX}_2^+ \in \mathsf{YSub}(f)$ for every $f \geq 1$; the rank-two CFG for B displayed below also shows that this ingredient is context-free. Thus the intrinsic rank-two phenomenon used here already occurs inside the trivial-typing baseline. The role of L_{intr} is to combine that known rule-rank obstruction with non-context-freeness, minimum fan-out two, and a separation from $\mathsf{YSub}(f)$ in a single finitely typed target.

Theorem 5.3 (A non-context-free intrinsic-rank-two typed witness)**.** *The language L_{intr} satisfies $L_{\mathrm{intr}} \in \mathsf{MCFL}(2,2) \setminus \mathrm{CFL}$, and, for every $f \geq 2$,*

$$L_{\mathrm{intr}} \in \mathcal{C}^{\mathrm{mcf}}_{f, h^{1,1}_{\Sigma_{\mathrm{intr}}}} \setminus \mathsf{YSub}(f).$$

Moreover, for every finite $f' \geq 1$, $L_{\mathrm{intr}} \notin \mathsf{MCFL}(f', 1)$. Consequently the minimum fan-out of L_{intr} is exactly two, and its minimum rule rank among all finite-fan-out MCFG presentations is exactly two. In particular, Theorem 4.9 applies to a non-context-free language that cannot be covered by Theorem 4.5 at any finite fan-out.

Proof. First we give a reduced good fan-out-two grammar of maximum rule rank two. Let B generate the nonempty balanced-count language by

$$\begin{aligned} B \to\ & \mathtt{a}B\mathtt{b}B \mid \mathtt{b}B\mathtt{a}B \mid \mathtt{a}B\mathtt{b} \mid \mathtt{ab}B \mid \mathtt{ab} \\ & \mid \mathtt{b}B\mathtt{a} \mid \mathtt{ba}B \mid \mathtt{ba}. \end{aligned}$$

The usual first-return induction on $|x|_{\mathtt{a}} - |x|_{\mathtt{b}}$ gives $L_B = \mathrm{MIX}_2^+$.

Let the fan-out-two nonterminal A have rules $A \to (\mathtt{u}, \mathtt{w}) \mid (\mathtt{u}\xi_1^1\mathtt{v}, \mathtt{w}\xi_1^2)(A)$. It derives exactly the tuples $\{(\mathtt{u}^n\mathtt{v}^{n-1}, \mathtt{w}^n) : n \geq 1\}$. For every $\alpha \in P_{\mathrm{intr}}$, add the start rule $S \to \alpha\#\xi_1^1\$\xi_2^1\mathtt{v}\xi_2^2(B, A)$. The resulting grammar generates exactly L_{intr}. It is reduced, λ-free, linear, nondeleting, and nonpermuting. It is also nonmerging: the two components of A are separated by the literal terminal $\mathtt{v}$ in every start rule. Hence it is a good fan-out-two grammar of maximum rule rank two.

The language is not context-free. Intersect it with the regular language $R := \mathtt{lpc\#ab\$}\,\mathtt{u}^+\mathtt{v}^+\mathtt{w}^+$. Then $L_{\mathrm{intr}} \cap R = \{\mathtt{lpc\#ab\$}\mathtt{u}^n\mathtt{v}^n\mathtt{w}^n : n \geq 1\}$. A homomorphism erasing the fixed prefix and fixing $\mathtt{u}, \mathtt{v}, \mathtt{w}$ maps this language onto $\{\mathtt{u}^n\mathtt{v}^n\mathtt{w}^n : n \geq 1\}$, which is not context-free. Closure of context-free languages under intersection with regular languages and homomorphism therefore shows that L_{intr} is not context-free. Since fan-out-one MCFGs are exactly context-free grammars in the present syntax, while the displayed grammar has fan-out two, the minimum fan-out is exactly two.

For typed membership, let

$$Q_{\mathrm{intr}} := P_{\mathrm{intr}} \,\#\, \{\mathtt{a}, \mathtt{b}\}^+ \,\$\, \mathtt{u}^+\mathtt{v}^+\mathtt{w}^+.$$

This language is strictly 2-local. One complete set of allowed adjacent pairs, including end markers, is

$$\begin{gathered}
\vdash \mathtt{l},\ \vdash \mathtt{r}, \quad \mathtt{lp},\ \mathtt{lq},\ \mathtt{rp}, \quad \mathtt{pc},\ \mathtt{pd},\ \mathtt{qc}, \\
\mathtt{c\#},\ \mathtt{d\#}, \quad \mathtt{\#a},\ \mathtt{\#b}, \quad \mathtt{aa},\ \mathtt{ab},\ \mathtt{ba},\ \mathtt{bb}, \\
\mathtt{a\$},\ \mathtt{b\$}, \quad \mathtt{\$u}, \quad \mathtt{uu},\ \mathtt{uv},\ \mathtt{vv},\ \mathtt{vw},\ \mathtt{ww}, \quad \mathtt{w} \dashv .
\end{gathered}$$

Define the homomorphism $\Phi_{\mathrm{intr}} : \Sigma_{\mathrm{intr}}^* \to \mathbb{Z}^3$ by

$$\begin{aligned}
&\Phi_{\mathrm{intr}}(\mathtt{a}) = (1,0,0), \quad \Phi_{\mathrm{intr}}(\mathtt{b}) = (-1,0,0), \\
&\Phi_{\mathrm{intr}}(\mathtt{u}) = (0,1,0), \quad \Phi_{\mathrm{intr}}(\mathtt{v}) = (0,-1,1), \quad \Phi_{\mathrm{intr}}(\mathtt{w}) = (0,0,-1),
\end{aligned}$$

and let every other terminal have image $(0,0,0)$. Then $L_{\mathrm{intr}} = Q_{\mathrm{intr}} \cap \Phi_{\mathrm{intr}}^{-1}(0)$. Indeed, the three coordinates impose respectively

$$|x|_{\mathtt{a}} = |x|_{\mathtt{b}}, \qquad |x|_{\mathtt{u}} = |x|_{\mathtt{v}}, \qquad |x|_{\mathtt{v}} = |x|_{\mathtt{w}}.$$

Lemma 5.2 and Proposition 2.10(ii) show that the displayed intersection is $(f, h^{1,1}_{\Sigma_{\mathrm{intr}}})$-tuple-substitutable for every $f \geq 1$. Together with the fan-out-two grammar above, this gives $L_{\mathrm{intr}} \in \mathcal{C}^{\mathrm{mcf}}_{f, h^{1,1}_{\Sigma_{\mathrm{intr}}}}$ for every $f \geq 2$.

The language is outside Yoshinaka's trivial-typing class already at arity one. Take $x := \mathtt{p}$, $y := \mathtt{q}$, and one-hole contexts $E := \mathtt{l}\Box_1\mathtt{c\#ab\$uvw}$, $F := \mathtt{r}\Box_1\mathtt{d\#ab\$uvw}$. Then $E[x],\ E[y],\ F[x] \in L_{\mathrm{intr}}$, $F[y] \notin L_{\mathrm{intr}}$. Thus the arity-one substitution implication fails, so $L_{\mathrm{intr}} \notin \mathsf{YSub}(f)$ for every $f \geq 1$. Notice again that the boundary typing distinguishes the offending one-letter factors: $h^{1,1}_{\Sigma_{\mathrm{intr}}}(\mathtt{p}) \neq h^{1,1}_{\Sigma_{\mathrm{intr}}}(\mathtt{q})$.

It remains to prove intrinsic rule rank. Put $\mathrm{MIX}_2 := \{x \in \{\mathtt{a}, \mathtt{b}\}^* : |x|_{\mathtt{a}} = |x|_{\mathtt{b}}\}$. Gallot calls an MCFG *non-branching* when every positive-rank rule has a single child nonterminal; in the present terminology this is exactly rule rank at most one, with no globally fixed fan-out bound [21, Definition 38]. Since every finite grammar has a finite maximum nonterminal arity, Gallot's non-branching MCFL class is $\bigcup_{f' < \infty} \mathsf{MCFL}(f', 1)$. Gallot's Theorem 29 identifies this class with the finite-index EDT0L languages, and Theorem 31 proves that MIX_2 is not finite-index EDT0L. Definition 38 explicitly permits a child variable to occur at most once, and Theorem 28 confirms that non-permuting non-erasing presentations generate the same non-branching class; thus the use of erasing terminal homomorphisms below stays within the formalism of the cited lower bound.

Equivalently, MIX_2 is not generated by any non-branching MCFG; Gallot's Propositions 5–6 give the underlying bounded-k-derivation contradiction [21, Theorem 29, Theorem 31, Propositions 5–6].

Bishop et al. later give a peer-reviewed expansion of the same obstruction in the closely corresponding R-MCFG formalism. Their Definition 6.1 uses *rank* for nonterminal arity (the present fan-out), while *non-branching* still means that a propagating rule has a single child nonterminal. Lemma 6.8 supplies the normal form; Definition 7.1 identifies the same balanced language, and Section 7 proves the obstruction, culminating in the stronger non-EDT0L statement of Theorem C [22, Definition 6.1, Lemma 6.8, Definition 7.1, Section 7, Theorem C]. This is corroborating evidence; Gallot's Theorems 29 and 31 remain the direct implication used here.

Now let π fix $\mathtt{a}, \mathtt{b}$ and erase every other terminal in Σ_{intr}. Then $\pi(L_{\mathrm{intr}}) = \mathrm{MIX}_2^+$. If L_{intr} had a fan-out-f' rule-rank-one presentation, applying π to the terminal material in its rule templates would preserve fan-out and rule rank and would give such a presentation of MIX_2^+. Empty terminal pieces created by erasure are permitted in the standard MCFG formalism. Adding a fresh start symbol with one unary identity link to the old start and one rank-zero rule generating ε would then give a fan-out-f' rule-rank-one presentation of MIX_2, contradicting Gallot's non-branching lower bound. Hence $L_{\mathrm{intr}} \notin \mathsf{MCFL}(f', 1)$ for every finite $f' \geq 1$. Together with the displayed rank-two grammar, this proves the claimed intrinsic rule rank. □

The construction is deliberately modular: the MIX_2^+ factor carries the obstruction to rule rank one, while the T_3^+ factor forces the target outside CFL and hence to minimum fan-out two. Thus the theorem establishes the coexistence of the two lower bounds in a single finitely typed target; the two bounds arise from separate factors rather than from an irreducible interaction between the resources.

The non-branching lower bound itself is prior work. What is specific to the present framework is that a strictly local wrapper and the $(1, 1)$ boundary typing place this target inside the typed class, while the displayed four-instance substitution implication keeps it outside Yoshinaka's trivial-typing baseline. The wrapper retains the prior non-branching obstruction, so the same target also has intrinsic rule rank two across all finite fan-outs.

6 Conclusion

We specialize Yoshinaka's positive-data signature framework [15] to fixed finite-semigroup typing. At bounded rule rank the qualitative reconstruction and polynomial-time updates are consequences of that framework, via the explicit substructure-grammar correspondence in Lemma 3.18. The uncapped-rank case retains a separate finite-sample argument. These qualitative facts do not determine the total length of characteristic data.

For total positive-data length, rank-one target rules admit a grammar-size polynomial for every fixed typing. At higher rank, boundary windows give a quadratic bound in grammar size and original thickness, while windows combined with arbitrary finite groups give a cubic bound. The local-group theorem (Appendix D) extends the latter to every typing with positive-image semigroup in **LG**. The corresponding pseudovariety-level obstruction uses fan-out-one lower-bound examples; it is not a per-typing characterization at arbitrary fan-out.

A complementary language-based result, Theorem 4.19, establishes polynomial characteristic-data length for abelian-filtered targets with bounded-rank good presentations, including when the finite control is outside **LG**. Proposition 4.17 isolates short representatives of target distribution classes as a sufficient condition; Proposition 4.15 instead gives a computable, presentation-sensitive rule-only certificate using grammar-generated representatives. These two constructions have different hypotheses and should not be conflated.

The boundary-typed non-context-free example has intrinsic fan-out and rule rank two (Theorem 5.3). Full yield-type refinement and empty-component elimination can also increase thickness exponentially (Appendix A).

Open Problem 6.1 (Original-thickness bounds beyond fan-out one). Fix $f \geq 2$ and a rule-rank bound $r \geq 2$. Does there exist a fixed finite-semigroup typing s and a family of reduced good fan-out-$\leq f$, rank-$\leq r$ grammars G_n of targets in $\mathcal{C}^{\mathrm{mcf}}_{f,s}$ for which $\|G_n\| \max\{1, \theta_{G_n}\}$ grows polynomially in n but every characteristic sample for $\mathcal{B}^{\leq r}_{f,s}$ has superpolynomial total positive length? The fan-out-one counterexample of the companion CFG paper [25] does not by itself answer this higher-arity question: membership in $\mathcal{C}^{\mathrm{mcf}}_{1,s}$ need not imply membership in $\mathcal{C}^{\mathrm{mcf}}_{f,s}$ for $f \geq 2$.

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Data availability

No empirical datasets were used. The source manuscript and scripts for reproducing finite illustrative checks are available in the public repository https://github.com/growupkuriyama-hub/Papers/tree/main/02_fixed-h-mcfg. The main branch contains the canonical working source for this manuscript. The general theorems are supported by mathematical proofs in this article, not by exhaustive computational testing.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

The author used ChatGPT (OpenAI) and Claude (Anthropic) during the research and manuscript-preparation process, including exploratory mathematical argument development, proof drafting and criticism, literature and consistency checks, and editing of text and LaTeX source. AI-generated suggestions and reviews are not treated as independent mathematical verification. The author takes responsibility for the mathematical claims, citations, and final content of this article.

A Details of the Good Normalization

The following bookkeeping records the fixed-parameter bounds used in Proposition 2.6; semantic correctness of the nondeleting and nonmerging conversions is standard [14, 30, 32].

Nondeleting conversion and component permutations. After trimming, implement the nondeleting construction (f3) in the proof of Seki et al.'s Lemma 2.2 [30, proof of Lemma 2.2, item (f3)] by retaining nonempty subsets I of output components. For $R : A \to \rho(B_1, \ldots, B_q)$ and such I, let J_j comprise exactly the child-j components used by the I-indexed outputs. Keep those outputs, reindex their variables, use $B_j^{J_j}$ for $J_j \neq \emptyset$, and omit a child for $J_j = \emptyset$. The new rule is nondeleting, and omitting children cannot increase rank. For reduced input, every omitted child is productive, so the projected rule can be lifted to a source rule: $L_{A^I} = \pi_I(L_A)$. Without prior trimming only $\pi_I(L_A) \subseteq L_{A^I}$ is automatic. This also establishes the quantitative bound directly: there are $O_f(1)$ projected copies of each nonterminal and $O_{f,r_{\max}}(1)$ copies of each rule. Kanazawa's existence statement [32, Section 2.1] does not by itself establish these size or thickness bounds. For nonpermutation, decorate a fan-out-d nonterminal by $\sigma \in \mathfrak{S}_d$. The parent permutation determines the child permutations, so there are at most $f!$ copies, no child is duplicated, and rank is preserved. Denote the result by G_{nd}.

Removing empty tuple components. For a fan-out-d nonterminal A, let

$$\mathsf{Pat}(A) := \{I \subseteq \{1, \ldots, d\} : I \text{ is the nonempty-component pattern of some } \vec{w} \in L_A(G_{\mathrm{nd}})\}.$$

Finite saturation computes these sets. For realizable child patterns, replace variables from absent components by ε, delete empty parent components, and omit all-empty children. There are at most 2^f patterns per nonterminal and $2^{fr_{\max}}$ choices per rule. Introduce A^I for each nonempty pattern I. Retained variables still occur once and in order, so nondeletion/nonpermutation, fan-out, and rank are preserved and the result is λ-free.

Induction on derivations gives L_{A^I} exactly as the projection of source tuples having pattern I; all-empty omitted children can be restored by terminal derivations realizing their all-empty component pattern. Since the start symbol has fan-out one, $S^{\{1\}}$ generates precisely the nonempty part of the original language. Projection shortens templates, giving $O_{f,r_{\max}}(\|G_{\mathrm{nd}}\|)$ size.

Nonmerging. Apply Yoshinaka's nonmerging conversion [14, Section 2.3, Lemma 1]. For the construction in the proof of the cited lemma, a merge-decorated nonterminal generates the *image* of its preceding tuple language under consecutive-component concatenation. The merge need not be injective: this image equality, not a claim about unique preimages, is the fact used in Proposition A.1. The construction preserves λ-freeness, nonpermutation, fan-out, and rule rank. For fixed fan-out and rank Yoshinaka's combined good-form conversion has $O(\|G\|)$ size [14, Section 2.3, after Example 2]; trimming can only decrease all bounds. This proves Proposition 2.6.

Thickness transfer for λ-free presentations

Proposition A.1 (Thickness under good-form conversion)**.** *Fix f and $r_{\max}$. Let G_{in} be a reduced λ-free f-MCFG of maximum rule rank at most $r_{\max}$, possibly with deleting rules, and suppose $L(G_{\mathrm{in}}) \cap \Sigma^+ \neq \emptyset$. The good normalization of Proposition 2.6 yields an equivalent reduced good grammar G with the same fan-out and rule-rank bounds such that*

$$\|G\| = O_{f,r_{\max}}(\|G_{\mathrm{in}}\|), \qquad t_G \leq t_{G_{\mathrm{in}}}, \qquad \theta_G = O_{f,r_{\max}}(\|G_{\mathrm{in}}\| \max\{1, t_{G_{\mathrm{in}}}\}).$$

Proof. We track the generated tuple languages through the normalization. For $\vec{w} = (w_1, \ldots, w_d)$ and nonempty $I \subseteq [d]$, let $\pi_I(\vec{w})$ retain the components indexed by I in increasing order. Write $\sigma\vec{w}$ for a component permutation and $\mu(\vec{w})$ for concatenation of consecutive blocks of components. Then

$$\|\pi_I(\vec{w})\|_1 \leq \|\vec{w}\|_1, \qquad \|\sigma\vec{w}\|_1 = \|\mu(\vec{w})\|_1 = \|\vec{w}\|_1.$$

Because G_{in} is λ-free, every component of every derived tuple is nonempty, by induction on derivations.

Nondeletion. For each nonterminal A and nonempty $I \subseteq [\text{fan}(A)]$, introduce A^I. Given $R : A \to \rho(B_1, \ldots, B_q)$, let J_j collect precisely the indices of the components of B_j occurring in the I-indexed output components of ρ. Keep those outputs, rename their variables and use $B_j^{J_j}$ for $J_j \neq \emptyset$; omit children with $J_j = \emptyset$. This produces a nondeleting rule with no larger fan-out or rank and with nonempty outputs. For arbitrary derived child tuples the identity

$$\pi_I(\rho(\vec{v}_1, \ldots, \vec{v}_q)) = \rho^I((\pi_{J_j}\vec{v}_j)_{J_j \neq \emptyset})$$

holds. Induction on derivations gives

$$L_{A^I} = \pi_I(L_A(G_{\text{in}})). \tag{*}$$

For the reverse inclusion, lift all retained child tuples using the induction hypothesis and supply a terminal derivation of each omitted child. This is possible because the source grammar was reduced. Thus deleting an entirely unused child does not change the projected tuple language.

Nonpermutation and empty components. For each $\sigma \in \mathfrak{S}_{\text{fan}(A)}$, decorate A by σ. The order in which a permuted parent template uses each child's variables determines the permutation of that child, since the grammar is linear and nondeleting. An induction on derivations identifies the new tuple language with σL_A, preserving total length, λ-freeness, fan-out and rank. Since all derived components are nonempty, the later empty-component-pattern elimination is the identity on this grammar.

Nonmerging. Use the construction in the proof of [14, Section 2.3, Lemma 1]. Decorate each nonterminal with a composition μ of its fan-out, representing concatenation of consecutive blocks. In the merged parent template, maximal adjacent runs of consecutive variables of a child determine that child's merge μ_j. Nonpermutation makes these runs compatible with the child's component order, while nondeletion ensures that they cover all its components. The resulting nonmerging template satisfies

$$\rho^\mu(\mu_1(\vec{v}_1), \ldots, \mu_q(\vec{v}_q)) = \mu(\rho(\vec{v}_1, \ldots, \vec{v}_q)).$$

Induction gives the *image equality* for every merge-decorated nonterminal. No inverse-image claim is needed, because the merge map need not be injective. Concatenation preserves total length, λ-freeness, fan-out and rank.

Consequently every final nonterminal X is associated with an original nonterminal A such that

$$L_X(G) = \mu(\sigma\pi_I(L_A(G_{\text{in}}))), \qquad \min_{\vec{u} \in L_X(G)} \|\vec{u}\|_1 \leq \min_{\vec{w} \in L_A(G_{\text{in}})} \|\vec{w}\|_1.$$

Trimming leaves the tuple languages of surviving nonterminals unchanged, so $t_G \leq t_{G_{\text{in}}}$. For fixed $f, r_{\max}$, the subset, permutation and merge decorations yield only fixed-parameter linearly many rule copies of no greater template size. Hence $\|G\| = O_{f,r_{\max}}(\|G_{\text{in}}\|)$. Finally Lemma 4.1 gives

$$\theta_G \leq 2\|G\| \max\{1, t_G\} = O_{f,r_{\max}}(\|G_{\text{in}}\| \max\{1, t_{G_{\text{in}}}\}).$$

□

Thus deleting rules do not by themselves obstruct polynomial thickness transfer from a reduced λ-free presentation; genuine empty components are the problematic step.

Proposition A.2 (Empty-component elimination can blow up thickness)**.** *For every $n \geq 1$ there is a reduced fan-out-two, rule-rank-two MCFG E_n of size $O(n)$ with $t_{E_n} \leq 2$ and $\theta_{E_n} = O(n)$ such that the empty-component elimination stage of Proposition 2.6 contains a retained nonterminal and a retained rule of thickness at least 2^n.*

Proof. Let E_n have fan-out-one S, fan-out-two $A_0, \ldots, A_n$, and

$$\begin{aligned} A_0 &\to (\mathtt{a}, \varepsilon), \\ A_i &\to (\varepsilon, \mathtt{c}) \mid (\xi_1^1\xi_2^1, \xi_1^2\xi_2^2)(A_{i-1}, A_{i-1}) \quad (1 \leq i \leq n), \\ S &\to (\xi_1^1 \mathtt{d}\, \xi_1^2)(A_n). \end{aligned}$$

It is reduced, nondeleting, nonpermuting, and nonmerging. Short side derivations give $t_{E_n} \leq 2$, and placing any chosen rule on one branch of the binary spine while completing siblings minimally gives $\theta_{E_n} = O(n)$.

If an A_i-tuple has nonempty first component and empty second component, the rule $(\varepsilon, \mathtt{c})$ cannot occur. Inductively the shortest such tuple is $(\mathtt{a}^{2^i}, \varepsilon)$. Thus pattern $\{1\}$ for A_n survives empty-component elimination, and the projected start rule $S \to \xi_1^1 \mathtt{d}\, (A_n^{\{1\}})$ can succeed only with a child of length at least 2^n. Hence the retained nonterminal and rule have thickness at least 2^n. □

This is the same obstruction exhibited by Proposition 4.4: finite state refinement can force exponentially long shortest witnesses inside one refined state.

B Windows combined with finite abelian groups

Semigroup convention (v82 migration). The group morphism $g : \Sigma^* \to A$ is retained as a computational monoid representation. The learning bias is the *positive* semigroup restriction of the window–group product, with a fresh identity adjoined only for evaluating empty template factors. By Lemma 3.7, nonempty component tests and the selected characteristic words are unchanged. The finite-group zero-sum argument uses g only as an auxiliary invariant.

This appendix extends the polynomial total-length theorem to finite-abelian-group refinements of a prefix–suffix window. The original Theorem 4.9, which does not involve a group, remains unchanged.

Lemma B.1 (Short tuples with prescribed abelian-group values)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG of fan-out at most f and rule rank at most $r_{\max} \geq 1$. Let $g : \Sigma^* \to A$ be a homomorphism into a fixed finite abelian group. Write $\mathsf{d}(H)$ for the maximal length of a zero-sum-free sequence over a finite abelian group H, so that $\mathsf{D}(H) = \mathsf{d}(H) + 1$ is its Davenport constant. For every $B \in V$ and every realizable vector $\mathbf{s} \in A^{\mathrm{fan}(B)}$, there is $\vec{u} \in L_B(G)$ with $g^{(\mathrm{fan}(B))}(\vec{u}) = \mathbf{s}$ and $\|\vec{u}\|_1 \leq \beta_g := (r_{\max} + 1)\bar{\theta}_G \max\{1, \mathsf{d}(A^f)\, |V||A|^f\}$.*

Proof. Choose a terminal derivation T from B whose root has g-vector $\mathbf{s}$, minimizing first the total yield length and then the number of derivation nodes. Label each node by its nonterminal C and the componentwise g-vector of its subtree yield. There are at most $n_g := |V||A|^f$ possible such labels.

If an ancestor and a proper descendant had the same label, replacing the former subtree by the latter would preserve the root g-vector: every MCFG rule computes the parent's group values from the values of its children. The total yield length cannot increase because the rules are nondeleting, and the number of nodes decreases. This contradicts the choice of T. Thus its depth is at most n_g.

Call a node *heavy* if the total length of its subtree yield exceeds $\bar{\theta}_G$. All ancestors of a heavy node are heavy by nondeletion. Suppose $v_1, \ldots, v_m$ are pairwise incomparable heavy nodes and

the root has fan-out $d \le f$. For each v_i labelled by C_i, choose a terminal derivation from C_i of total length at most $\bar{\theta}_G$, which exists by Lemma 4.6(ii). Replacing the subtree at v_i by this short derivation strictly decreases the total yield length.

Because the grammar is linear and nondeleting, the output components of mutually incomparable subtrees contribute disjoint terminal blocks to the root components, possibly in different orders. In an abelian group, multiplication is additive and the order of those blocks is immaterial. Therefore there are vectors $\delta_i \in A^d$ such that simultaneous replacement of any subset J of the v_i changes the root g-vector by $\sum_{i\in J} \delta_i$. If $m > \mathsf{d}(A^d)$, the sequence $\delta_1 \cdots \delta_m$ over A^d has a nonempty zero-sum subsequence $(\delta_i)_{i\in J}$, so simultaneous replacement at the nodes v_i, $i \in J$, preserves the root vector while strictly shortening its yield. This contradicts minimality. Every heavy antichain consequently has at most $\mathsf{d}(A^d) \le \mathsf{d}(A^f)$ nodes; the inequality holds because A^d embeds in A^f.

If the root is light, its length is at most $\bar{\theta}_G$. Otherwise the heavy nodes form an ancestor-closed tree. Its leaves are a heavy antichain, and every heavy node belongs to a path to one of these leaves. There are therefore at most $\mathsf{d}(A^f)\, n_g$ heavy nodes. Every explicit terminal factor written by a rule is of total length at most $\bar{\theta}_G$ by Lemma 4.6(i). Every light child of a heavy node contributes at most $\bar{\theta}_G$ additional terminals, and a heavy node has at most $r_{\max}$ children. Counting the heavy nodes and their light children gives the asserted bound. The case $|A| = 1$ is covered by the light-root alternative. □

Remark B.2 (Size of the zero-sum constant). The prefix-sum pigeonhole argument gives $\mathsf{d}(A^f) \le |A|^f - 1$, so Lemma B.1 is never weaker than the bound obtained from consecutive partial sums. It is usually much stronger. Write $A \cong C_{n_1} \oplus \cdots \oplus C_{n_s}$ and $n = \exp(A) \ge 2$. Then

$$f \sum_{i=1}^{s} (n_i - 1) \;\le\; \mathsf{d}(A^f) \;\le\; (n-1) + n \log \frac{|A|^f}{n},$$

where the lower bound is attained by a basis sequence and the upper bound is due to van Emde Boas and Kruyswijk [34, Proposition 5.1.8 and Theorem 5.5.5]. If A is a p-group, the lower bound is an equality [34, Theorem 5.5.9] (Olson's theorem). Hence $\mathsf{d}(A^f) = O(\exp(A)\, f \log |A|)$ grows linearly in the fan-out bound, whereas $|A|^f - 1$ grows exponentially; for $A = C_2$ the antichain bound drops from $2^f - 1$ to f.

The value $\mathsf{d}(A^d)$ is also the limit of the antichain step as stated. That step uses only the fact that the difference vectors δ_i have no nonempty zero-sum subsequence, and a zero-sum-free sequence of length $\mathsf{d}(A^d)$ exists. We do not claim that a grammar realizing such an extremal configuration exists.

Lemma B.3 (Short canonical tuples for window–abelian typing)**.** *Assume the hypotheses of Lemma B.1 and fix $k, \ell \ge 0$ and a finite typing $h : \Sigma^* \to M$ satisfying $h \preceq (h_\Sigma^{k,\ell}, g)$. Let $\widetilde{G}_0$ be the trimmed full h-refinement of G, and let n_{typ} be its number of non-start nonterminals. For each such nonterminal X one may choose a canonical tuple $\omega(X)$ with $\|\omega(X)\|_1 \le f(k+\ell+1)n_{\mathrm{typ}}\,(\bar{\theta}_G + r_{\max}\beta_g)$.*

Proof. Choose, among the typed terminal derivations from X, one with the fewest nodes. A repeated typed nonterminal on a path could be eliminated while preserving the root h-type, so each path has at most n_{typ} nodes. As in the proof of Lemma 4.8, mark in each root component the positions that determine its $h_\Sigma^{k,\ell}$-type: all positions if the component is shorter than the window threshold, otherwise the first k and last ℓ positions, with one position marked when $k = \ell = 0$. The ancestor paths to all marks form a skeleton with at most $f(k+\ell+1)n_{\mathrm{typ}}$ nodes.

Each child subtree omitted from this skeleton lies wholly inside the unmarked interior of every root component to which it contributes. Record its nonterminal C and its actual componentwise

g-value $\mathbf{s}$. Replace it by a derivation from C with the same $\mathbf{s}$ and length at most β_g, as supplied by Lemma B.1. Goodness keeps all replacement components nonempty. No protected boundary position changes, and each long root component remains long, so its window type is unchanged. Moreover, each root component is a concatenation of the explicit terminal material and the child components; each replaced child component retains its g-value, so the root g-vector is also unchanged, without requiring a commutativity argument at this stage.

The product $(h_\Sigma^{k,\ell}, g)$ and hence the coarser h-type of the root are unchanged. Forget the intermediate typed labels and uniquely lift the new underlying G-derivation back to the h-refinement using Proposition 3.6(ii). Because X is retained, the resulting root derivation survives trimming. Each skeleton node writes at most $\bar{\theta}_G$ terminals and has at most $r_{\max}$ omitted children. The displayed length estimate follows. □

Theorem B.4 (Polynomial data for window–abelian typing)**.** *Fix $f, r_{\max}, k, \ell, \Sigma$, a finite abelian group A, a homomorphism $g : \Sigma^* \to A$, and a finite typing $h \preceq (h_\Sigma^{k,\ell}, g)$. For every $L \in \mathcal{C}_{f,h}^{\mathrm{mcf}}$ and reduced good fan-out-$\le f$, rank-$\le r_{\max}$ grammar G with $L(G) = L \cap \Sigma^+$, the capped reconstruction operator $\mathcal{B}_{f,h}^{\le r_{\max}}$ has a characteristic sample satisfying*

$$|\mathsf{W}(\widetilde{G}_0)| = O(\|G\|), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},k,\ell,h,g,\Sigma}(\|G\|^3 \bar{\theta}_G).$$

Proof. The fixed finite typing gives $n_{\mathrm{typ}} \le |V||M|^f = O(\|G\|)$. Lemma B.1 gives $\beta_g = O(\|G\|\bar{\theta}_G)$. By Lemma B.3, the selected canonical tuple lengths are $O(\|G\|^2\bar{\theta}_G)$. Lemma 4.7 therefore bounds each characteristic witness word by $O(\|G\|^2\bar{\theta}_G)$; Proposition 3.11 bounds their number by $O(\|G\|)$. The ε datum, if any, adds only a constant to the encoded total. □

Remark B.5. Commutativity is used in the zero-sum antichain step of Lemma B.1. At fan-out one, subtrees contribute single intervals, and the companion CFG paper treats arbitrary finite groups. For $f \ge 2$ and nonabelian groups, the pieces from different subtrees can interleave in different root components. The zero-sum argument in this appendix does not cover that case. The separate order-homogenization proof in Appendix C does cover arbitrary finite groups, including nonabelian groups.

C Windows combined with arbitrary finite groups

Semigroup convention (v82 migration). The finite group A and its monoid morphism g are auxiliary to the shortening proof. The semantic typing is the positive-image semigroup restriction of the window–group product; the canonical fresh-identity completion evaluates template gaps. The transfer lemma 3.7 preserves all nonempty tuple comparisons and the statement's polynomial exponent.

Here we remove commutativity from the finite-group hypothesis in Appendix B. The learning operator and the window-only bound are unchanged. All groups in this appendix are fixed and finite. The constants below are deliberately very large: the result concerns polynomial dependence on the *original* grammar size and rule thickness, not a practical uniform bound in the order of the group.

Lemma C.1 (A homogeneous block-order lemma)**.** *Fix positive integers e, n with $n \ge 3$. For each $t \in [e]$ and $i \in [n]$, let (t, i) be a distinct point on a line, with a total left-to-right order. Suppose that (i) the order of (t, i) and (u, i) is independent of i; and (ii) for each ordered pair (t, u), including $t = u$, the order of (t, i) and (u, j) for $i < j$ is independent of the choice of $i < j$. Then the types at the points have a partition into classes, each occupying a consecutive region of the line. Every class*

has one direction (increasing or decreasing in i); inside its region the points occur in n successive chunks*, one per rank in that direction. Each chunk consists of exactly one point of each type in the class (in a fixed within-chunk order).*

Proof. By (ii) with $t = u$, each type's points are monotone. Consider distinct types t, u. Write X for $(t, i) < (u, j)$ when $i < j$, and Y for $(u, i) < (t, j)$ when $i < j$; both are uniform by (ii). If both types increase, $X \wedge Y$ puts all points of rank i before all points of rank $j > i$. The cases $X \wedge \neg Y$ and $\neg X \wedge Y$ put all t-points before all u-points, or conversely. The remaining case $\neg X \wedge \neg Y$ would give $(t, 1) < (t, 2) < (u, 1) < (u, 2) < (t, 1)$, a contradiction. If t increases and u decreases, $X \wedge Y$ gives $(t, 2) < (u, 3) < (u, 1) < (t, 2)$, while $\neg X \wedge \neg Y$ gives $(t, 3) < (u, 2) < (t, 1) < (t, 3)$. For $X \wedge \neg Y$, all off-diagonal comparisons put t to the left of u; a contrary equal-rank comparison would give $(t, 1) < (u, 2) < (u, 1) < (t, 1)$, contrary to (i). Thus the types are separated. The case $\neg X \wedge Y$ is symmetric. Reverse indices to cover the two-decreasing case. Thus any two types are either *separated*, or have the same direction and are *interleaved by rank*.

Interleaving is transitive. Indeed, if t interleaves with u and u with v, all three have the same direction. Reverse indices to assume increasing. If t and v were separated with t to the left, we would have $(v, 1) < (u, 2) < (t, 3) < (v, 1)$; if v were to the left of t, we would instead have $(t, 1) < (u, 2) < (v, 3) < (t, 1)$. Hence interleaving, with equality, partitions the types into classes. A type outside a class is separated from each type inside it, and must lie on the same side of all its types: otherwise two interleaved types would themselves be separated. The class regions are therefore consecutive. The rank comparisons give successive chunks. Finally the within-chunk order is independent of rank by (i). □

Lemma C.2 (Simultaneous finite-group interval shortening)**.** *Let A be any finite group. Suppose n indices each contribute at most f disjoint blocks to at most f ordered output components, and each block has a prescribed replacement block. A selected index replaces* all *its blocks simultaneously. The other material is fixed. If the individual block destinations and within-index orders are constant across the indices, and all cross-index block order comparisons are constant for $i < j$, then $n \geq \max\{3, |A|^f\}$ implies that a nonempty consecutive interval of indices can be replaced without changing any output component's value in A.*

Proof. Apply Lemma C.1 separately to the block types going to each output component. Across all output components there are $q \leq f$ consecutive class regions. Consider such a region K, and enumerate its chunks $1, \ldots, n$ from left to right. Include all unchanged material between the blocks in the region. Let P_p be the original group value of its prefix ending after chunk p, and Q_p the value of the same prefix when chunks $1, \ldots, p$ have been replaced; set $P_0 = Q_0 = 1_A$. Write $\kappa_p^K = P_p Q_p^{-1}$.

Replacing just chunks $p + 1, \ldots, p'$ gives the following *ordered* value for the entire region:

$$P_p \, Q_p^{-1} Q_{p'} \, P_{p'}^{-1} P_n.$$

Indeed, $Q_p^{-1} Q_{p'}$ is exactly the replacement value of the segment between the two prefix boundaries, and $P_{p'}^{-1} P_n$ is its unchanged suffix. Consequently, the region retains its original value P_n whenever $\kappa_p^K = \kappa_{p'}^K$. This computation uses inverses, but never commutes factors.

If class K increases with the original index order, its chunks for indices $(a, b]$ occupy $(a, b]$. If it decreases, they occupy $(n - b, n - a]$. For $j = 0, \ldots, n$ define a state in A^q by taking κ_j^K from each increasing class and κ_{n-j}^K from each decreasing class. There are $n + 1 > |A|^q$ such states, so two, at indices $a < b$, coincide. Replacing precisely indices $a + 1, \ldots, b$ then preserves each class region's value. The rest of every output component is unchanged, so all output group values are preserved. □

Lemma C.3 (Polynomial order homogenization)**.** *Let $N \geq 3$, and define $m_0 = N$ and $m_{s+1} = (m_s - 1)^2 + 1$, so that $m_k \leq N^{2^k}$. Let T be a finite set of point types, each assigned to one of several lines. For every index $i \in [n]$ and $t \in T$ let $p_t(i)$ be a point on the line of t, the points on each line being distinct. Let Π be the set of ordered pairs (t, u), including $t = u$, of types on a common line, and put $k = |\Pi|$. If $n \geq m_k$, there is a set $I \subseteq [n]$ of N indices such that, for every $(t, u) \in \Pi$, the truth value of $p_t(i) < p_u(j)$ is the same for all $i < j$ in I.*

Proof. Treat the pairs one at a time, the pairs (t, t) first. Each step passes from a current index set of size at least $(m-1)^2 + 1$ to a subset of size at least m. Homogeneity already obtained, and monotonicity of each type, persist in subsets.

For (t, t), homogeneity means that p_t is monotone on the subset. The Erdős–Szekeres theorem gives a monotone subsequence of length m.

For $t \neq u$, both p_t and p_u are already monotone. Relabel the current indices $1, \ldots, m'$ in order, with $m' \geq (m-1)^2 + 1$. For $i < j$ put $X(i, j) = [p_t(i) < p_u(j)]$. Each row $X(i, \cdot)$ is monotone in j, and each column $X(\cdot, j)$ is monotone in i. Replacing X by $1 - X$ if necessary, we may assume that the rows go from 0 to 1. This does not change which sets are homogeneous.

If the columns go from 1 to 0, put $J(i) = \min\{j > i : X(i, j) = 1\}$, or $J(i) = \infty$ if there is no such j. Then J is nondecreasing. Set $i_1 = 1$ and $i_{s+1} = J(i_s)$ while this is at most m'. For $a < b$ we have $i_b \geq J(i_a)$, so $X(i_a, i_b) = 1$. If the chain has at least m terms, its first m terms suffice. If it has fewer than m terms, its at most $m - 1$ terms cut $[m']$ into at most $m - 1$ intervals $[i_s, i_{s+1})$ and $[i_L, m']$, and one interval has at least $m'/(m-1) > m - 1$ indices. Inside $[i_s, i_{s+1})$, every $i < j$ satisfies $j < i_{s+1} = J(i_s) \leq J(i)$, so $X(i, j) = 0$. The same holds in the last interval, where $J \equiv \infty$.

If the columns go from 0 to 1, the set of pairs with $X = 1$ is closed upward in both coordinates. Let κ be the least index with $X(\kappa, \kappa + 1) = 1$, or $\kappa = m'$ if there is none. For $\kappa \leq i < j$ we have $X(i, j) \geq X(\kappa, \kappa + 1) = 1$. For $i < j \leq \kappa$ we have $X(i, j) \leq X(j - 1, j) = 0$. One of $[1, \kappa]$ and $[\kappa, m']$ has at least $(m' + 1)/2 \geq m$ indices, because $(m-1)^2 + 1 \geq 2m - 1$ for $m \geq 3$.

After all k pairs the remaining set has at least $m_0 = N$ indices. Finally, $m_{s+1} \leq m_s^2$ gives $m_k \leq N^{2^k}$. □

Lemma C.4 (Uniform shortening of heavy antichains)**.** *Set*

$$N_{A,f} = \max\{3, |A|^f\}, \quad U_f = f^{f+1} f!, \quad M_{A,f} = U_f \, N_{A,f}^{2^{f^2}}.$$

Let G be reduced and good, and let $g : \Sigma^* \to A$ be any homomorphism into a finite group. In a terminal derivation minimizing first total yield length and then node count subject to its prescribed componentwise g-value, every antichain of subtrees of total yield length $> \bar{\theta}_G$ has fewer than $M_{A,f}$ nodes.

Proof. Every child component passes through the linear, nondeleting templates to exactly one contiguous block in a root component. Blocks of incomparable subtrees are disjoint. Fix for each node of a heavy antichain a replacement derivation of the *same untyped* nonterminal whose yield has length at most $\bar{\theta}_G$; such a derivation exists by Lemma 4.6(ii). Its group values may differ from the original values.

An antichain node has at most f blocks. Its *unary pattern* records the number of blocks, their root-component destinations, and their order within any shared root component. There are at most U_f such patterns. An antichain of at least $M_{A,f}$ nodes therefore contains at least $N_{A,f}^{2^{f^2}}$ nodes with one common unary pattern. Enumerate them in any fixed order. The block types of the common pattern are points of at most f types, placed on the root components as lines. There are at most f^2 ordered pairs of types sharing a root component, including $t = u$. Since $m_k \leq N^{2^k} \leq N^{2^{f^2}}$ for

$k \le f^2$, Lemma C.3 yields $N = N_{A,f}$ indices with constant unary pattern and constant cross-index comparisons between (t,i) and (u,j) for all $i < j$.

Within each root component their blocks meet the hypotheses of Lemma C.1. Lemma C.2 therefore supplies a nonempty interval of these selected indices whose simultaneous replacement preserves the *whole root group vector*. Every replaced subtree was heavy but is replaced by one of length at most $\bar{\theta}_G$, so the resulting valid derivation has strictly smaller root yield length. This contradicts its choice. □

Theorem C.5 (Short tuples with prescribed arbitrary finite-group values)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG of fan-out at most f and maximum rule rank $r_{\max} \ge 1$. Let $g : \Sigma^* \to A$ be a homomorphism into* any *finite group A. For every $B \in V$ and every realizable $\mathbf{s} \in A^{\mathrm{fan}(B)}$ there is $\vec{u} \in L_B(G)$ with $g^{(\mathrm{fan}(B))}(\vec{u}) = \mathbf{s}$ and*

$$\|\vec{u}\|_1 \le \beta_g^{\mathrm{grp}} := (r_{\max}+1)\bar{\theta}_G \max\{1, (M_{A,f}-1)|V||A|^f\}.$$

In particular $\beta_g^{\mathrm{grp}} = O_{A,f,r_{\max}}(\|G\|\bar{\theta}_G)$.

Proof. Choose a terminal derivation from B having the prescribed root vector, lexicographically minimizing (total length, node count). Label each node by its nonterminal and componentwise group value. An ancestor and a proper descendant cannot have the same label: replacing the ancestor by the descendant would preserve every ancestral group vector (by the ordered evaluation of rule templates), would not increase total length (by nondeletion), and would decrease node count. Hence each root-to-leaf path has at most $n_g = |V||A|^f$ nodes.

Call a node heavy when its subtree's total yield length exceeds $\bar{\theta}_G$. Heavy nodes are ancestor-closed. By Lemma C.4, their leaves number at most $M_{A,f} - 1$; all heavy nodes lie on paths to those leaves, so their number is at most $(M_{A,f}-1)n_g$. If the root is light, its yield already has length at most $\bar{\theta}_G$. Otherwise, each heavy rule contributes at most $\bar{\theta}_G$ explicit terminal symbols, and each of its at most $r_{\max}$ light child subtrees contributes at most $\bar{\theta}_G$. Counting the heavy nodes and their maximal light children gives the displayed upper bound. □

Lemma C.6 (Short canonical tuples for window–arbitrary-group typing)**.** *Under the hypotheses of Theorem C.5, fix $k, \ell \ge 0$ and a finite typing h satisfying $h \preceq (h_\Sigma^{k,\ell}, g)$. Let $\widetilde{G}_0$ be the trimmed full h-refinement of G, and let n_{typ} denote its number of non-start nonterminals. Every non-start X of $\widetilde{G}_0$ admits a canonical tuple $\omega(X)$ of total length at most*

$$f(k+\ell+1)n_{\mathrm{typ}}(\bar{\theta}_G + r_{\max}\beta_g^{\mathrm{grp}}).$$

Proof. Choose a terminal derivation from X minimizing the number of nodes. Along a root-to-leaf path, a repeated typed nonterminal could be bypassed, preserving the root h-type and decreasing the number of nodes; thus each path has at most n_{typ} nodes. As in Lemma 4.8, mark the root positions determining the $h_\Sigma^{k,\ell}$-value: all positions in a short component, the first k and last ℓ positions in a long component, and one position when $k = \ell = 0$. The union of ancestor paths to the at most $f(k+\ell+1)$ marks is a skeleton with at most $f(k+\ell+1)n_{\mathrm{typ}}$ nodes.

For each child subtree omitted from this skeleton, record its underlying nonterminal B and the componentwise group vector actually realized. Theorem C.5 supplies a replacement derivation from B of length at most β_g^{grp} and the *same group value in each component*. Since G is good, all replacement components are nonempty. No protected position changes, and each root component that was long remains long; hence the window value is preserved. The ordered evaluation of rule templates preserves every root group value because each replaced child component retains its group value; no commutativity is used. Consequently the product $(h_\Sigma^{k,\ell}, g)$ and the coarser root h-type

are unchanged. The underlying G-derivation lifts to the trimmed full refinement with root X by Proposition 3.6(ii).

Each skeleton rule contributes at most $\bar{\theta}_G$ explicit terminals and at most $r_{\max}$ omitted children, each contributing at most β_g^{grp} terminals. The displayed bound follows. □

Theorem C.7 (Polynomial characteristic data for window–finite-group typing)**.** *Fix $f, r_{\max}, k, \ell, \Sigma$, any finite group A, a homomorphism $g : \Sigma^* \to A$, and a fixed finite typing h satisfying $h \preceq (h_\Sigma^{k,\ell}, g)$. For every target $L \in \mathcal{C}_{f,h}^{\mathrm{mcf}}$ and reduced good fan-out-$\le f$, rank-$\le r_{\max}$ grammar G with $L(G) = L \cap \Sigma^+$, the capped reconstruction operator $\mathcal{B}_{f,h}^{\le r_{\max}}$ has a characteristic sample such that*

$$|\mathsf{W}(\widetilde{G}_0)| = O(\|G\|), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},k,\ell,h,g,\Sigma}(\|G\|^3 \bar{\theta}_G).$$

Proof. By Lemma C.6, canonical tuples in the trimmed refinement have length at most

$$b_{\mathrm{can}} \le f(k+\ell+1) n_{\mathrm{typ}} (\bar{\theta}_G + r_{\max} \beta_g^{\mathrm{grp}}) = O(\|G\|^2 \bar{\theta}_G),$$

since $n_{\mathrm{typ}} \le |V||M|^f = O(\|G\|)$ and the typing and rank bounds are fixed. Lemma 4.7 bounds each characteristic word by $2\bar{\theta}_G + r_{\max} b_{\mathrm{can}} = O(\|G\|^2 \bar{\theta}_G)$, and Proposition 3.11 bounds the number of words by $O(\|G\|)$. The optional empty-word witness adds only a constant to the encoded total. □

Remark C.8. This strengthening does not assert an original-thickness polynomial for arbitrary finite *monoid* typings. For fixed f the constant $M_{A,f} = U_f \max\{3, |A|^f\}^{2^{f^2}}$ is polynomial in $|A|$. It replaces the multicolor Ramsey bound $U_f\, c_f^{c_f(N_{A,f}-1)+1}$ with $c_f = 2^{f^2}$ used in earlier versions, which is exponential in $|A|^f$. The dependence on f remains doubly exponential. The abelian argument of Appendix B gives a much smaller constant, linear in f, when A is commutative (Remark B.2).

D Local-group typings and the semilattice boundary

Semigroup convention (v82 migration). Throughout this appendix **LG** is tested on the *positive image* $S = h(\Sigma^+)$ of the typing, not on the canonical completion $S^\sharp$. Auxiliary group-valued maps and the displayed monoid morphisms are implementations, whose restrictions to nonempty words define the semantic bias. Lemma 3.7 justifies using the former presentation in the quantitative proofs.

The window–group typings of Appendices B and C have positive images in the pseudovariety join $\mathbf{LI} \vee \mathbf{G}$ of finite locally trivial semigroups and finite groups [25, Lemma 7.4]. Here we extend the polynomial original-thickness bound to the strictly larger pseudovariety **LG** of *local groups*. This is the class of finite semigroups S in which every local monoid eSe, with e idempotent, is a group. Classically, $\mathbf{LG} = \mathbf{G} * \mathbf{D}$ [38, 39]. The equivalence between being a local group, having a nilpotent extension of the minimal ideal, and containing no two-element semilattice is *known* [37, Proposition 2.3]. Theorem D.4 recalls that equivalence and adds the explicit sliding-factor group typing characterization used for the MCFG quantitative bound. In particular, **LG** is exactly the class of finite semigroups containing no copy of the two-element semilattice $U_1 = \{1, \zeta\}$ ($\zeta^2 = \zeta$). That semilattice is the positive image of the typing behind the exponential lower bound of [25]. At the level of pseudovarieties, the semilattice is therefore the only obstruction (Corollary D.9).

Throughout this appendix Σ is nonempty, $k \ge 1$, and A is a finite group. For a word w with $|w| \ge k+1$ let $\sigma_k(w)$ be the sequence of its factors of length $k+1$, ordered by their starting positions.

Definition D.1 (Local-group typing)**.** Let $g : \Sigma^{k+1} \to A$ be any map. For $|w| \geq 2k$ put

$$\gamma_{k,g}(w) := g(F_1)g(F_2)\cdots g(F_{|w|-k}), \qquad \sigma_k(w) = (F_1, \ldots, F_{|w|-k}),$$

the ordered product in A, and put $\gamma_{k,g}(w) := 1_A$ for $|w| < 2k$. The *local-group typing* is

$$\lambda_{k,g}(w) := \big(h_\Sigma^{k,k}(w), \gamma_{k,g}(w)\big).$$

A finite typing h is *local-group-determined* if $h \preceq \lambda_{k,g}$ for some k, A and g.

Recall from Definition 2.4 that, for $\ell = k$, words of length $< 2k$ are *short* and recorded exactly, while a *long* word w is recorded by $\langle \mathrm{pref}_k(w), \mathrm{suf}_k(w)\rangle$. For $s, p \in \Sigma^k$ write $J_g(s,p) := \prod_{i=1}^{k} g((sp)[i..i+k])$. This is the ordered product over the k factors of length $k+1$ of sp that start in s.

Lemma D.2 (Local-group typings are morphisms)**.** *The kernel of $\lambda_{k,g}$ is a congruence of finite index on Σ^*. Hence $\lambda_{k,g}$ is a homomorphism onto a finite monoid $M_{k,g}$. For long u, v,*

$$\lambda_{k,g}(uv) = \big(\langle \mathrm{pref}_k\, u, \mathrm{suf}_k\, v\rangle,\ \gamma_{k,g}(u)\, J_g(\mathrm{suf}_k\, u, \mathrm{pref}_k\, v)\, \gamma_{k,g}(v)\big).$$

Proof. The window component is a morphism by Definition 2.4. For long u, v, the factors of length $k+1$ of uv are, in order: those of u, then the k factors starting in the last k positions of u, then those of v. The middle factors lie inside $\mathrm{suf}_k(u)\,\mathrm{pref}_k(v)$, which gives the displayed formula.

If u is short and v is long, the factors of uv that start in u lie inside $u\,\mathrm{pref}_k(v)$, and the remaining factors are those of v. Thus $\gamma_{k,g}(uv)$ is determined by u, $\mathrm{pref}_k(v)$ and $\gamma_{k,g}(v)$. The case of long u and short v is symmetric. If both are short, uv is determined exactly.

In every case $\lambda_{k,g}(uv)$ is a function of $\lambda_{k,g}(u)$ and $\lambda_{k,g}(v)$. The image has at most $|\Sigma|^{<2k} + |\Sigma|^{2k}|A|$ elements. □

Lemma D.3 (Block decomposition)**.** *Let $w = x_0\beta_1 x_1 \cdots \beta_q x_q$ be a factorization of a word into designated blocks β_j and fixed words x_j. Let $w' = x_0\beta'_1 x_1 \cdots \beta'_q x_q$, where $h_\Sigma^{k,k}(\beta'_j) = h_\Sigma^{k,k}(\beta_j)$ for every j. Then $h_\Sigma^{k,k}(w') = h_\Sigma^{k,k}(w)$. If w is long, there are $c_0, \ldots, c_q \in A$, depending only on the x_j and the window types of the β_j, such that*

$$\gamma_{k,g}(w) = c_0\, \gamma_{k,g}(\beta_1)\, c_1 \cdots \gamma_{k,g}(\beta_q)\, c_q, \qquad \gamma_{k,g}(w') = c_0\, \gamma_{k,g}(\beta'_1)\, c_1 \cdots \gamma_{k,g}(\beta'_q)\, c_q.$$

Proof. A short block is recorded exactly, so $\beta'_j = \beta_j$; absorb such blocks into the fixed words. Every remaining block is long, and its replacement is long with the same first and last k letters. Hence w' is long exactly when w is, and its first and last k letters are unchanged. This gives the window claim.

Classify the factors F of length $k+1$ of w. If F lies inside a long block β_j, it is one of the factors of β_j. Otherwise F meets each long block it touches only in a prefix of length at most k or a suffix of length at most k. If it met both ends of β_j, then $|\beta_j| \leq k-1$, so β_j would be short. Hence every factor of the second kind is spelled by the fixed words and the windows $\mathrm{pref}_k(\beta_j)$, $\mathrm{suf}_k(\beta_j)$. The same factors, in the same order, occur in w'. Ordering all factors by starting position, the factors inside β_j form one consecutive run contributing $\gamma_{k,g}(\beta_j)$. Let c_j be the ordered product of the remaining factors between the runs of β_j and β_{j+1}. □

Theorem D.4 (Characterizations of local-group positive images)**.** *For a finite typing $h : \Sigma^* \to M$ put $S := h(\Sigma^+)$. The following are equivalent.*

(i) $S \in \mathbf{LG}$*; that is,* eSe *is a group for every idempotent* e *of* S*.*

(ii) S *has no subsemigroup isomorphic to* U_1*.*

(iii) *Every idempotent of* S *lies in the minimal ideal of* S*.*

(iv) h *is local-group-determined. One may take* $k = |S|$*.*

Proof. The implications among (i)–(iii) follow from the standard characterization [37, Proposition 2.3]; we include their short semigroup argument to make the reduction self-contained. The additional equivalence with (iv) is the windowed-factor formulation required for learning.

(i)⇒(ii). Let $\{e, f\} \cong U_1$ with $ef = fe = f \neq e$. Then $f \in eSe$ is an idempotent of a group whose identity is e, so $f = e$, a contradiction.

(ii)⇒(iii). Let K be the minimal ideal of S; it contains an idempotent. For an idempotent e of S choose $s \in K$, and let $x := (ese)^\omega$. This is an idempotent in $K \cap eSe$, so $ex = xe = x$. If $x \neq e$ then $\{e, x\} \cong U_1$. Hence $e = x \in K$.

(iii)⇒(i). The minimal ideal K of a finite semigroup is completely simple [40]. For an idempotent $e \in K$ we have $eSe \subseteq K$, so $eSe = eKe$. In a completely simple semigroup this is the $\mathcal{H}$-class of e, a group.

(iii)⇒(iv). By the Rees theorem [40] write $K = \mathcal{M}(H; I, \Lambda; P)$, with product $(i, a, \lambda)(j, b, \mu) = (i, ap_{\lambda j}b, \mu)$. Put $n := |S|$.

Long products lie in K*.* The Rees quotient S/K has no idempotent other than 0, since all idempotents of S lie in K. Let $s_1, \dots, s_m \in S$ with $m := |S/K| \leq n$, and consider the prefix products in S/K. If none is 0, two of them coincide, say $p_i = p_j$ with $i < j$. Then $p_i = p_i x = p_i x^\omega = 0$ for $x = s_{i+1} \cdots s_j$, a contradiction. Hence $h(w) \in K$ whenever $|w| \geq n$.

Right action. Fix $i_0 \in I$ and for $\lambda \in \Lambda$ let $e_\lambda := (i_0, p^{-1}_{\lambda i_0}, \lambda)$, an idempotent. For $x = (i, a, \lambda) \in K$ we have $xe_\lambda = x$. Hence, for $s \in S$, $xs = x(e_\lambda s)$. Writing $e_\lambda s = (j, c, \mu) \in K$ gives

$$(i, a, \lambda)\, s = (i,\ a\,\rho(\lambda, s),\ \mu(\lambda, s)), \qquad \rho(\lambda, s) := p_{\lambda j} c,$$

where $\rho(\lambda, s) \in H$ and $\mu(\lambda, s) \in \Lambda$ depend only on (λ, s). Dually, the Λ-coordinate of $h(w)$ for $|w| \geq n$ is that of $h(\mathrm{suf}_n w)$, and the I-coordinate is that of $h(\mathrm{pref}_n w)$.

Window factorization. Let $w = a_1 \cdots a_m$ with $m \geq 2n$, and let λ_j be the Λ-coordinate of $h(a_{j-n+1} \cdots a_j)$ for $j \geq n$. Iterating the right action gives

$$h(w) = \Big(i,\ a \prod_{j=n}^{m-1} \rho(\lambda_j, h(a_{j+1})),\ \lambda_m\Big), \qquad h(a_1 \cdots a_n) = (i, a, \lambda_n).$$

The j-th factor of the product depends only on the factor $a_{j-n+1} \cdots a_{j+1}$ of length $n + 1$. Define $g(F) := \rho(\lambda(h(F[1..n])), h(F[n + 1]))$ for $F \in \Sigma^{n+1}$. Then the product is $\gamma_{n,g}(w)$. So $h(w)$ is determined by $\mathrm{pref}_n w$, $\mathrm{suf}_n w$ and $\gamma_{n,g}(w)$. Words of length $< 2n$ are recorded exactly by $h^{n,n}_\Sigma$, and ε is the only word of its $\lambda_{n,g}$-class. Thus $h \preceq \lambda_{n,g}$ with $A = H$.

(iv)⇒(i). $\mathbf{LG}$ is closed under homomorphic images, so it suffices to show $T := \lambda_{k,g}(\Sigma^+) \in \mathbf{LG}$. A short element is never idempotent: the square of a short word is either longer and short, or long. Let $e = (\langle p, s\rangle, \gamma_e)$ be an idempotent. By Lemma D.2, $\gamma_e J_g(s, p)\gamma_e = \gamma_e$. For every $x \in T$, exe has window $\langle p, s\rangle$, because e is long on both sides. Hence eTe lies in $H_e := \{(\langle p, s\rangle, \alpha) : \alpha \in A\}$. The product on H_e is $\alpha * \beta = \alpha J_g(s, p)\beta$. The map $\alpha \mapsto \alpha J_g(s, p)$ is an isomorphism from $(H_e, *)$ onto A. Thus eTe is a subsemigroup of a finite group, hence a group. □

Proposition D.5 (Window–group typings are local-group typings, strictly). (i) *If $h \preceq (h_\Sigma^{k,\ell}, g)$ for a homomorphism g into a finite group, then $h \preceq \lambda_{K,g'}$ with $K = \max\{k,\ell,1\}$ and $g'(F) := g(\text{last letter of } F)$. Hence* $\mathbf{LI} \vee \mathbf{G} \subseteq \mathbf{LG}$.

(ii) *Let $\Sigma = \{\mathtt{a},\mathtt{b},\mathtt{c}\}$. Let $h_{\mathtt{ab}} := \lambda_{1,g}$ with $g(\mathtt{ab}) = 1 \in \mathbb{Z}_2$ and $g(F) = 0$ otherwise; it records the parity of the number of occurrences of $\mathtt{ab}$, together with the window. Then $h_{\mathtt{ab}}$ is not window–group-determined. Hence $h_{\mathtt{ab}}(\Sigma^+) \in \mathbf{LG} \setminus (\mathbf{LI} \vee \mathbf{G})$.*

Proof. (i) For $|w| \geq 2K$, $g(w) = g(\mathrm{pref}_K w)\,\gamma_{K,g'}(w)$. Shorter words are recorded exactly. The inclusion of pseudovarieties follows from Theorem D.4 and the companion characterization of $\mathbf{LI} \vee \mathbf{G}$ by window–group typings [25, Lemma 7.4].

(ii) Suppose $h_{\mathtt{ab}} \preceq (h_\Sigma^{k,\ell}, g')$ with $g' : \Sigma^* \to A'$. Let $K = \max\{k,\ell,1\}$ and let q be an even multiple of the exponent of A'. Put $u = \mathtt{a}^K(\mathtt{abc})^q\mathtt{a}^K$ and $v = \mathtt{a}^K(\mathtt{bac})^q\mathtt{a}^K$. Both are long, with prefix $\mathtt{a}^k$ and suffix $\mathtt{a}^\ell$, and $g'(u) = g'(\mathtt{a})^{2K} = g'(v)$. But u contains q occurrences of $\mathtt{ab}$, while v contains exactly one, at $\mathtt{a}^K\mathtt{b}$. So $h_{\mathtt{ab}}(u) \neq h_{\mathtt{ab}}(v)$, a contradiction. Membership in $\mathbf{LG}$ follows from Theorem D.4. Non-membership in $\mathbf{LI} \vee \mathbf{G}$ follows from the companion characterization. □

Lemma D.6 (Short tuples with prescribed local-group type). *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG of fan-out at most f and rule rank at most $r_{\max} \geq 1$, and let $\lambda = \lambda_{k,g}$. Let $b_{\mathrm{w}} := f(2k+1)n_{\mathrm{w}}(r_{\max}+1)\bar{\theta}_G$, where $n_{\mathrm{w}} \leq |V||M_{k,k}|^f$ is the number of non-start typed nonterminals of the trimmed $h_\Sigma^{k,k}$-refinement. Put $n_\lambda := |V||M_{k,g}|^f$, and let $M_{A,f}$ be the constant of Lemma C.4. For every $B \in V$ and every realizable $\mathbf{t} \in M_{k,g}^{\mathrm{fan}(B)}$ there is $\vec{u} \in L_B(G)$ with $\lambda^{(\mathrm{fan}(B))}(\vec{u}) = \mathbf{t}$ and*

$$\|\vec{u}\|_1 \leq \beta_\lambda := (r_{\max}+1)\, b_{\mathrm{w}} \max\{1, (M_{A,f}-1)\, n_\lambda\}.$$

If A is abelian, $M_{A,f} - 1$ may be replaced by $\mathtt{d}(A^f)$.

Proof. Choose a terminal derivation T from B with root type $\mathbf{t}$ that minimizes first total yield length and then node count. Label each node by its nonterminal and the λ-vector of its yield. As in Lemma B.1, no label repeats on a path, because λ is a morphism (Lemma D.2) and rule templates evaluate homomorphically. Hence every path has at most n_λ nodes.

Call a node *heavy* if its yield has total length greater than b_{w}; heavy nodes are ancestor-closed. Let v be a heavy node labelled by C, and let $\boldsymbol{\omega}$ be the $h_\Sigma^{k,k}$-vector of its yield. Since G is reduced, T can be inserted into a successful derivation. Hence $C_{\boldsymbol{\omega}}$ is a retained typed nonterminal of the trimmed $h_\Sigma^{k,k}$-refinement, and Lemma 4.8 supplies a derivation from C whose yield has the same window vector and total length at most b_{w}. Fix this replacement for every heavy node.

Consider a heavy antichain $v_1, \ldots, v_m$. Each v_i contributes at most f disjoint blocks to the root components. Replacing the subtrees at any set J of indices yields a valid derivation of strictly smaller total length. By Lemma D.3, applied to each root component, the replacement leaves every root window type unchanged. It also changes each long root component's γ-value only through an ordered product in which each block contributes its own γ-value between fixed constants. This is exactly the setting of Lemma C.2, whose proof uses only this ordered-product structure. The proof of Lemma C.4 therefore applies verbatim, with threshold b_{w} in place of $\bar{\theta}_G$, with the window-preserving replacements fixed above, and with γ-values in place of g-values. It uses unary patterns, Lemma C.3 and Lemma C.2. If $m \geq M_{A,f}$, some nonempty set of replacements preserves the whole root λ-vector, contradicting minimality. For abelian A the zero-sum argument of Lemma B.1 applies to the differences $\gamma(\beta') - \gamma(\beta)$ and gives the bound $\mathtt{d}(A^f)$.

Hence the heavy nodes number at most $(M_{A,f}-1)n_\lambda$. Each heavy node writes at most $\bar{\theta}_G \leq b_{\mathrm{w}}$ terminals and has at most $r_{\max}$ light children, each of yield length at most b_{w}. If the root is light, its length is at most b_{w}. □

Theorem D.7 (Polynomial data for local-group typings)**.** *Fix $f, r_{\max}, \Sigma$ and a finite typing h with $h(\Sigma^+) \in \mathbf{LG}$. For every $L \in \mathcal{C}^{\mathrm{mcf}}_{f,h}$ and reduced good fan-out-$\le f$, rank-$\le r_{\max}$ grammar G with $L(G) = L \cap \Sigma^+$, the capped operator $\mathcal{B}^{\le r_{\max}}_{f,h}$ has a characteristic sample with*

$$|\mathsf{W}(\widetilde{G}_0)| = O(\|G\|), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},h,\Sigma}(\|G\|^3 \bar{\theta}_G).$$

Proof. By Theorem D.4, $h \preceq \lambda_{k,g}$ for some k, A, g depending only on h. Let $X = B_{\mathbf{s}}$ be a non-start typed nonterminal of the trimmed h-refinement and $\vec{u} \in L_X$. Lemma D.6, applied to the λ-vector of $\vec{u}$, gives $\vec{u}' \in L_B(G)$ with the same λ-vector, hence the same h-vector, and $\|\vec{u}'\|_1 \le \beta_\lambda$. By Proposition 3.6(ii) its derivation lifts to root X, and it survives trimming as in Lemma 4.8. So canonical tuples of length at most $\beta_\lambda = O(\|G\|^2 \bar{\theta}_G)$ exist. Here we use $b_{\mathrm{w}} = O(\|G\| \bar{\theta}_G)$ and $n_\lambda = O(\|G\|)$. Lemma 4.7 bounds every witness word by $2\bar{\theta}_G + r_{\max}\beta_\lambda$, and Proposition 3.11 bounds their number by $O(\|G\|)$. □

Theorem D.7 contains the window and window–group theorems as special cases (Proposition D.5(i)), with the same exponent three in the group case. The intersection lemma below makes the lower-bound side precise.

Lemma D.8 (Type filters preserve substitutability)**.** *If L is (f,h)-tuple-substitutable and $F \subseteq M$, then $L \cap h^{-1}(F)$ is (f,h)-tuple-substitutable.*

Proof. Let $\vec{x}, \vec{y}$ have equal h-vectors and share a context E in $L' := L \cap h^{-1}(F)$. They share E in L, so they have the same L-distribution. If $E'[\vec{x}] \in L'$, then $E'[\vec{y}] \in L$ and $h(E'[\vec{y}]) = h(E'[\vec{x}]) \in F$, so $E'[\vec{y}] \in L'$. □

Corollary D.9 (Pseudovariety dichotomy)**.** *Let $\mathbf{V}$ be a pseudovariety of finite semigroups.*

(i) *If $\mathbf{V} \subseteq \mathbf{LG}$, then for every alphabet, every typing h with $h(\Sigma^+) \in \mathbf{V}$, and all fixed $f, r_{\max}$, characteristic data of total length $O(\|G\|^3 \bar{\theta}_G)$ exist as in Theorem D.7.*

(ii) *If $\mathbf{V} \not\subseteq \mathbf{LG}$, then $U_1 \in \mathbf{V}$. At fan-out one there is a typing h_{U} with $h_{\mathrm{U}}(\Sigma^+) = U_1$ and targets with $\|G\|, \theta_G = O(n)$, every characteristic sample of which for $\mathcal{B}^{\le 2}_{1,h_{\mathrm{U}}}$ contains a word of length $3 \cdot 2^n - 2$. The same holds for every set-driven operator admitting characteristic samples throughout $\mathcal{C}^{\mathrm{mcf}}_{1,h_{\mathrm{U}}}$.*

Proof. (i) is Theorem D.7. For (ii), choose $S \in \mathbf{V} \setminus \mathbf{LG}$. By Theorem D.4, S has a subsemigroup isomorphic to U_1, and pseudovarieties are closed under subsemigroups.

Take the doubling family L_n, the typing h_{U} and the long type-1 word w^*_n of [25, Proposition 6.4]. There L_n is shown to be Clark–Eyraud substitutable, hence $(1, h_{\mathrm{U}})$-tuple-substitutable. By Lemma D.8, $L^-_n := L_n \cap h_{\mathrm{U}}^{-1}(\zeta) = L_n \setminus \{w^*_n\}$ is as well. If $K \subseteq L_n$ omits w^*_n, then $K \subseteq L^-_n$, and Proposition 3.4 gives $L(\mathcal{B}^{\le 2}_{1,h_{\mathrm{U}}}(K)) \subseteq L^-_n$. Thus every characteristic sample contains w^*_n. The set-driven statement is the distinguishing-pair argument of [25, Proposition 6.4(iv)] for the pair L_n, L^-_n. □

Remark D.10 (Scope of the dichotomy). (i) Part (i) holds at every fixed fan-out. Part (ii) is established only at fan-out one. The doubling family does not lift directly: L_2 is not $(2, h_{\mathrm{U}})$-tuple-substitutable. The type-1 pairs $([, \mathtt{a}])$ and $([\mathtt{a},])$ share the context $[[\mathtt{aa}]\,\square_1\,\mathtt{a}\,\square_2\,]$ but have different distributions. This was checked exhaustively for $n \le 3$. Under the finer typing recording each component's set of letters, which still has semilattice image, the corresponding failure appears at $n = 3$. Whether part (ii) holds at fan-out $f \ge 2$ is the pseudovariety form of Open Problem 6.1.

(ii) The dichotomy concerns pseudovarieties. For an individual typing h with $h(\Sigma^+) \notin \mathbf{LG}$, the class $\mathcal{C}^{\mathrm{mcf}}_{1,h}$ need not contain a coded copy of the doubling family. Whether every such typing forces superpolynomial characteristic data remains open.

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